

Rewards, Credit Access, and Inequality in the Credit Card Market

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Abstract

This paper examines the distributional consequences of credit cards by accounting for their dual role as payment instruments and sources of unsecured credit. Existing studies typically frame these concerns around rebates earned net of interest charges and fee, which fails to capture the welfare gains from credit access. Using microdata from the Bank of Canada's Digital Wallet and Payment Trends Survey, I estimate a structural model of consumer payment choice and show that ignoring the value of credit access overstates the Gini coefficient of welfare gains from credit cards by 46%. Moreover, the Gini coefficient rises by 6% in a counterfactual that requires cardholders to repay their balances in full. This result implies that access to revolving credit has progressive distributional effects, and contradicts the perception of credit card debt as exploitative.

1 Introduction

As credit cards become increasingly widespread, so too have concerns about their effects on inequality. Existing literature typically frames this issue around the distribution of net rewards, defined as rebates earned net of interest charges and fees. Viewed through this lens, cardholders who carry outstanding balances (i.e., *Revolvers*) appear substantially worse off than *Convenience users*, who repay their balances in full and avoid interest charges. Because *Revolvers* are disproportionately lower-income cardholders, this disparity has fueled concerns about the regressive distributional effects of credit cards.

Rather than relying on net rewards, which primarily capture the value of credit cards as payment instruments (i.e., the payment function), this paper evaluates inequality in the welfare gains from credit cards using consumer surplus. In particular, I define consumer surplus as the compensating variation required to hold utility constant in a counterfactual world without credit cards. Because cash-based alternatives lack unsecured lending features (i.e., the credit function), this approach captures both the payment and credit functions of credit cards.

My primary contribution is showing that net rewards offer an incomplete measure of cardholder welfare because they capture the financial costs of borrowing while omitting the utility of credit access. By ignoring this liquidity channel, the conventional focus on net rewards systematically understates the benefits of credit cards for *Revolvers*, thereby overstating the regressive effects of credit card.

The empirical analysis uses the Bank of Canada’s Digital Wallet and Payment Trends Survey (DWPT). The survey reports consumers’ payment behavior, revolving status, expenditures, and the exact credit cards they hold. By matching these cards to data on rewards, fees, and interest rates, I estimate consumer preferences and calculate individual-level consumer surplus.

Results indicate that credit cards generate sizable but unequally distributed welfare gains. Back-of-the envelope calculations suggest that credit cards generate over CAD 9.1 billion per year in consumer surplus, approximately 9% of all social security benefits distributed by the Canadian government. These gains concentrate among high-income *Convenience users* resulting in a Gini coefficient of 0.58. While substantial, inequality measured using consumer surplus remains significantly lower than that implied by net

rewards, which have a Gini coefficient of 0.8. This comparison suggests that narrow focus on nominal differences in net rewards overstates the true extent of inequality by roughly 46%. Two mechanisms explain the divergence between these welfare measures.

First, descriptive evidence suggests that *Revolvers* rely on revolving credit to partially finance their spending. This allows these cardholders to maintain higher expenditures than their *Convenience users* despite earning lower incomes. Whether this debt-financed spending reflects deliberate consumption smoothing (Telyukova, 2013; Dao, 2025) or behavioral biases (Han, 2023; Agarwal et al., 2026), it generates consumption utility that net rewards fail to capture.

Second, net rewards abstract from heterogeneity in how consumers value card features. Consistent with prior work, I find that *Convenience users* receive positive net rewards averaging CAD 192 per year, whereas *Revolvers* incur average annual losses of CAD 672 (Agarwal et al., 2023). This disparity explains the high level of inequality in net rewards. Nevertheless, the demand estimates show that *Revolvers* place less value on reward features than *Convenience users*. Accounting for this preference heterogeneity further narrows the welfare gap between the two groups.

The counterfactual analysis explore the “tug-of-war” between the payment and credit functions. Specifically, I benchmark the baseline Gini coefficient against two counterfactuals. The first eliminates reward programs, removing a central component of the payment function. The second eliminates the credit function by requiring cardholders to always repay their balances in full.

Allocative inequality in consumer surplus, as measured by Gini coefficient, decreases by 12% under the no-reward counterfactual. This finding indicates that rebate programs are in fact regressive as established by the literature (Felt et al., 2023; Egan et al., 2026). This decline in inequality, nonetheless, comes at the cost of substantially lower surplus among *Convenience users*, whose surplus declines by as much as 30%.

By contrast, the Gini coefficient rises by 6% under the no-borrowing counterfactual. For *Revolvers*, much of the value of credit cards derives from access to liquidity rather than from rewards or other payment conveniences. Thus, removing the credit function reduces their surplus by up to 38%, and worsens inequality. This finding challenges the conventional view that credit card borrowing is inherently regressive.

Under this view, issuers extract interest revenue from financially naive

borrowers to subsidize rewards for more sophisticated cardholders (Heidhues and Kőszegi, 2010; Agarwal et al., 2023). My analysis instead shows that access to revolving credit can have progressive welfare effects by providing liquidity to lower-income cardholders. Evaluating credit cards solely through net rewards therefore obscures the opposing distributional effects of their payment and credit functions.

1.1 Literature Reviews

This paper contributes to the literature on the redistributive effects of credit cards. Studies that identify interest charges as the primary source of reward funding characterize the main transfer as occurring from *Revolvers*, who pay interest on outstanding balances, to *Convenience users*, who earn rewards while avoiding interest charges. Since credit card borrowing is often associated with poor financial management skills, this strain of literature suggests that the main margin of transfers is from financial naive consumers to their more sophisticated counterparts (Heidhues and Kőszegi, 2010; Agarwal et al., 2023).

By contrast, the payments literature emphasizes interchange fees as the primary funding channel for rewards. Although these fees are nominally charged to merchants, they are often passed through to consumers through higher retail prices (Huynh, Nicholls, and Shcherbakov, 2022; Wang, 2023). Because merchants generally do not differentiate prices by payment method, these costs are borne by all consumers, though not necessarily equally (Egan et al., 2026). This generates transfers from cash and debit card users to credit card users.

Unlike existing studies that directly measure cross-subsidization across consumer groups (Schuh, 2010; Felt et al., 2023; Agarwal et al., 2023; Egan et al., 2026), this paper focuses on allocative inequality in the welfare gains from credit cards. The objective is therefore not to identify the underlying financial transfers, but to quantify how unequally the resulting welfare gains are distributed across cardholders. For instance, while interchange-fee alters retail prices and implicitly shapes consumer surplus through observed spending, I do not attempt to quantify the underlying transfers across groups due to this mechanism.

The paper also shows that the two transfer margins emphasized in the literature are not cleanly separable. *Revolvers* use cash and debit cards more intensively than *Convenience users* and may therefore bear a disproportion-

ate share of interchange-fee pass-through. The distinction between transfers from *Revolvers* to *Convenience users* and transfers from non-credit-card payments to credit card payments is consequently less clear than the literature often implies. This distinction begs the question of why consumers differ in their demand for credit cards. While existing studies largely frame card use around the payment function (Koulayev et al., 2016; Arango, Huynh, and Sabetti, 2011), this paper shows that a substantial share of *Revolvers*' card use reflects demand for credit. Since revolving credit is costly, *Revolvers* use credit cards less intensively than *Convenience users*, despite deriving substantial liquidity value from them. Consequently, net rewards alone do not fully capture their welfare gains from credit cards.

On the theoretical front, this paper integrates an explicit model of payment choice into a framework that captures both the payment and credit functions of credit cards (Fulford and Schuh, 2024). My main theoretical contribution is to develop a static framework for evaluating welfare from an inherently dynamic borrowing decision. The main insight is that conditional on the observed levels of spending, which themselves are equilibrium objects of the dynamic problem, consumer choices between credit cards and other payment methods are static. The welfare effects of credit cards can therefore be recovered from the surplus generated by this static payment choice, together with the share of spending financed by new borrowing and its associated cost, without estimating the dynamic problem.

In that sense, my approach is also related to the literature using sufficient-statistic approach for welfare assessment (Egan et al., 2026). A major limitation of my approach is that it restricts the range of counterfactuals that can be evaluated. Since welfare is assessed conditional on observed expenditure, the framework is not well suited to counterfactuals that substantially alter credit demand and, consequently, spending levels.

2 Data

2.1 The Digital Wallet And Payment Trend Survey

‘Data source: The main data source for this paper is the Digital Wallet and Payment Trend Survey (DWPT), a cross-sectional study commissioned by the Bank of Canada and conducted by IPSOS. The survey provides a comprehensive overview of the Canadian payment market, offering detailed insights

into credit card ownership, transaction patterns, and borrowing behaviors. The key variations for this analysis include credit card usage, measured by both absolute monthly spending (CAD) and as a share of total expenditures (i.e., share of wallet), respondents’ revolving status and outstanding balances. The survey also contains a rich set of demographic and financial variables, including income, asset holdings, liabilities, and self-reported financial comfort. As the DWPT is a recall-based survey, the dataset does not offer detailed transaction-level information, such as merchant identity or point-of-sale transaction values.

A feature unique to the DWPT is that the survey reports the exact names of credit cards owned by each respondent (e.g., TD First Class Visa Travel Infinite). This allows me to match the specific credit card with its characteristics and estimate consumer preferences for these features. While other datasets also contain information on card features, they are often limited to indicators of availability such as whether consumers are collecting rewards (Ching and Hayashi, 2010; Henry, Shimoda, and Rusu, 2024), or subjective attitudinal measures already incorporating consumer preferences (Koulayev et al., 2016; Huynh, Nicholls, and Shcherbakov, 2022). As such, existing studies relying on these datasets usually offer rather coarse insights into consumer preferences.

Although credit report (e.g., Equifax, TransUnion) or banks’ administrative data may sometimes come with specific card names, these datasets only capture the portion of expenditure spent via credit cards. As such, they provide an incomplete view of consumers’ total spending, making it difficult to identify the substitution between different payment methods (Han, 2023; Fulford and Schuh, 2024).¹

Sample Selection: My analysis bases on the 2021 and 2022 surveys. I restrict the sample to respondents with positive credit card usage to focus on the welfare of cardholders. Given that 90% of Canadian 18 years and older own at least credit cards, the highest ownership rates worldwide (Demirguc-Kunt et al., 2022), this restriction is unlikely to materially affect the final result. The final sample consists of 10309 respondents equally distributed between the two sampling periods. The selected sample is representative of all cardholders in Canada (Dao, 2025). Table 1 reports the main summary statistics.

¹Not all credit reports contain card names. TransUnion data from the Bank of Canada, for example, can only identify the payment network and card issuers (e.g. TD Visa)

Card Characteristics: I constructed objective measures of card characteristics by mapping the specific credit cards to data from third-party platforms (e.g., creditcardgenius.ca) and official issuing bank websites. Because approximately 80% of credit card spending and borrowing is conducted through a single credit card, I focus on the characteristics of each respondent’s primary credit card. The final dataset contains 220 different credit cards detailing the type of rewards collected (i.e., cashback versus reward points), reward earning rates as a percentage of transaction volume, scope of insurance coverages, interest rate and annual fees. To facilitate comparison between cashback and point-based rebates, I convert the former into cashback-equivalent rates by multiplying the average points earned per dollar by the average value per point. Further details on the construction of this variable are provided in Appendix 10.1.

Definition of Revolvers: Following the literature, I define as *Revolvers* any cardholders who report to carry positive outstanding balances (Grodzicki and Koulayev, 2019; Fulford and Schuh, 2023; Allen, Boutros, and Guttman-Kenney, 2024). Conversely, *Convenience users* are cardholders who do not carry any outstanding balances. Under this definition, 21% of cardholders are classified as *Revolvers*. The reported revolving rate is consistently with other survey evidence for the Canadian market (Henry, Shimoda, and Rusu, 2024) but is less than half the rate observed in administrative credit report data (Xiao, 2024; Allen, Boutros, and Guttman-Kenney, 2024). This discrepancy is not specific to the DWPT, but instead reflects a well-documented divergence between survey-based and administrative for both the Canadian (Dao, 2025) and US markets (Zinman, 2009; Brown et al., 2011).

In a companion paper (Dao, 2025), I show that underreporting of revolving rate in surveys likely reflects differences in the types of revolving behavior captured across data sources. More specifically, surveys primarily capture persistent revolving behaviors, whereas credit reports capture both persistent and short-term *Revolvers*. In fact, 77% of *Revolvers*, as identified in the DWPT, report to “usually repay less than the full balance”.² Since these consumers borrow on a consistent basis, their behaviors likely reflect deliberate choices rather than only behavioral biases or a lack of financial sophistication, as is often suggested in the literature (Agarwal et al., 2023). Section 3.2 provides additional support for the rational interpretation of re-

²Redefining *Revolvers* as cardholders who **usually** repay less than the full balance” produces a similar revolving rate of 22%

volving behaviors.

The assertion that the DWPT captures predominantly persistent *Revolvers* finds further empirical support in the distribution of the debt-to-consumption ratio (Figure 1). With a mean outstanding balance exceeding \$4,000, the average *Revolver* carries a debt load nearly three times the size of their monthly consumption expenditure. Because new credit card borrowing in a given month is mechanically bounded above by total card spending, a first-time or transitory borrower must exhibit a debt-to-consumption ratio of at most one. For the 45% of *Revolvers* who maintain a debt-to-consumption ratio well above one, this implies that these users are carrying substantial balances from previous months while also accumulating new charges. This ongoing overlap of past and current debt indicates that revolving behavior, as captured in the DWPT, is a highly persistent financial state.

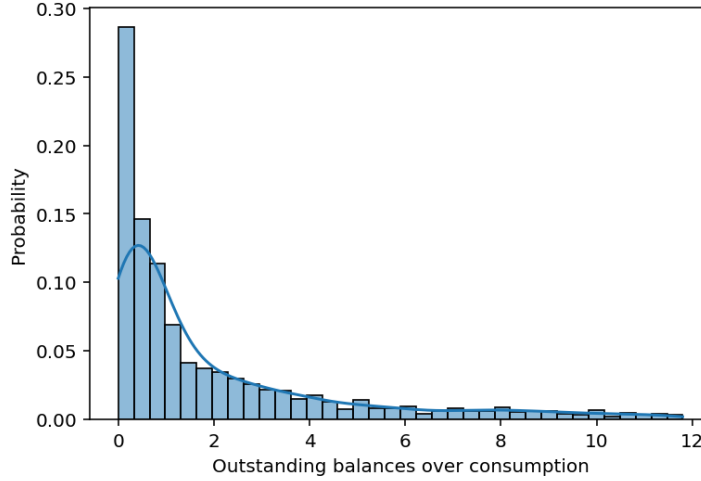
Table 1: Summary statistics of selected sample.

DWPT definition: respondents carrying positive amount of outstanding balances

Alternative definition: respondents who pay less than the entire balance are classified as *Revolvers*

Variable	Mean	Std	Median	75th	80th	90th	95th
No. of credit cards :	1.8	.77	2	2	3	3	3
% of expenses (Share of Wallet) :							
Cash	9.7	15.1	5	10	10	25	45
Debit card	21.6	27.8	10	40	50	70	80
Credit card	61.9	32.6	75	90	95	99	100
Other methods	6.8	13.6	0	10	10	20	40
Expenses by method (\$) :							
Cash	331.4	1397.2	87.5	235.3	300.0	600.0	1200.0
Debit card	794.4	8081.0	133.3	600.0	833.3	1750.0	3000.0
Credit card	1391.1	2318.1	950.0	1885.0	2000.0	3000.0	4000.0
Other methods	331.5	2608.1	0.0	175.0	250.0	640.0	1233.8
Monthly expense (\$)	2688	8579	1666	3000	3333	5000	7000
Revolving balance (\$)	895.5	3264	0	0	116.7	2000	5500
<i>Share of Revolvers (%)</i>							
DWPT definition	21.67						
Alternative definition	22.26						
Demographics:							
Age	50.6	16.3	52	64	66	72	75
Share with Bachelor degree (%)	46.4						
Share of Married (%)	.62						
Pretax Income (CAD/year)	84,277	49,756	80,000	100,000	125,000	150,000	200,000
Gender (Male =1)	.49						
Observations	10309						

Figure 1: Distribution of debt to consumption ratio - Revolver only



	Mean	Stdev	p25	p50	p75
Outstanding balance (CAD)	4037	5851	500	1800	5000
Debt-to-consumption ratio	2.7	5.17	0.28	0.88	2.85

3 Industry Background and Descriptive Evidence

3.1 Credit Card Market In Canada

Canada features a vibrant credit card market, with more than 250 credit cards offering a wide range of rewards and fee structures. Available options range from basic no-annual-fee products with reward earn rates of approximately 0.5%–1%, to premium cards with annual fees as high as CAD 600. These premium products can offer reward rates of up to 3%, along with benefits such as airport lounge access and extensive insurance coverage.³

Beyond the reward earn rate, cardholders can also select the type of rebates they wish to collect, typically choosing between cashback and points-based programs. Cashback cards provide a straightforward rebate as a percentage of transaction value. On the other hand, point-based programs allow cardholders to flexibly redeem their points across various reward categories,

³Examples include the TD Aeroplan Visa Infinite Privilege and American Express Platinum Card.

each of which implies a different monetary value per point. Table 11 provides an example of such a redemption scheme. This flexibility adds another dimension of heterogeneity to the market, as the actual value of points is contingent upon the individual’s ability to take advantage of high-value redemption channels.

Table 2: Value of TD Reward Points

Source: FinlyWealth Points Calculator

Redemption Option	Avg. Value
Expedia® For TD	0.5¢ / pt
Travel	0.44¢ / pt
Statement Credits	0.25¢ / pt
Gift Cards	0.25¢ / pt
Merchandise	0.225¢ / pt

3.2 Descriptive Facts

This section documents some empirical facts that motivate the need to move beyond net rewards as a sufficient measure of the redistributive effects of credit cards.

First, *Revolvers* earn substantially lower net rewards than *Convenience users*, and in many cases incur negative net rewards after interest payments and annual fees. This pattern is consistent with previous findings (Agarwal et al., 2023) and provides a useful benchmark for interpreting the subsequent descriptive facts. Second, *Revolvers* use credit cards not only as payment instruments, but also as a source of liquidity to finance current consumption. Third, payment choices vary systematically by revolving status: *Revolvers* rely more heavily on debit cards, use credit cards less intensively, and respond less strongly to credit card rewards.

Fact 2 and 3 show that *Convenience users* and *Revolvers* differ along multiple dimensions of credit card demand. Whereas *Convenience users* primarily value credit cards as payment instruments that provide rewards and other card features, *Revolvers* also value their liquidity function (Fact 2). At the same time, *Revolvers*’ weaker responsiveness to rewards suggests substantial variations in preferences for credit card features (Fact 3). As existing works often abstracts away from these dimensions (Felt et al., 2023;

Agarwal et al., 2023; Egan et al., 2026), Fact 2 and 3 highlight the need for a welfare framework that accounts for both consumer heterogeneity and the dual payment-credit function of credit cards.

While survey data may suffer from measurement errors, a companion paper provides extensive robustness checks to address these concerns, including alternative definitions of revolving status and a series of balance tests (Dao, 2025). The baseline descriptive facts documented below remain robust to these considerations.

Fact 1: Revolvers Earn Negative Net Rewards.

Following the literature (Agarwal et al., 2023), I calculate the net monetary benefits of credit cards as:

$$\begin{aligned} \text{Net rewards}(\$/\text{month}) = & \text{Reward rate} \times \text{Credit Card spending} \\ & - \text{Monthly interest rate} \times \text{Revolving balance} \\ & - \frac{\text{Annual fee}}{12}. \end{aligned} \tag{1}$$

Here, the reward rate denotes the rebate earning rate as a percentage of spending, while the monthly interest rate is computed as

$$\text{Monthly interest rate} = \frac{\text{Advertised annual rate}}{365} \times 30.5.$$

Finally, credit card spending and revolving balance refer to average monthly card spending and the balance on which interest is incurred, respectively. Because the DWPT reports both of these quantities at a monthly frequency, annual fees are also expressed on a monthly basis by dividing them by 12.

Figure 2 plots the distribution of monthly average Net rewards collected by *Convenience users* and *Revolvers*. On average, *Convenience users* gain roughly \$17/month from their card usage, whereas the average *Revolvers* lose \$60/month. Only 15% of *Revolvers* in our sample earn positive Net rewards versus 85% for *Convenience users*. This gap is largely mechanical, given that *Revolvers* carry substantial balances at an annual rate of roughly 21%.

This pattern is not unique to Canada, and has also been documented for the U.S (Agarwal et al., 2023). Since *Revolvers* tend to have lower educational attainment and lower income than their non-revolving counterparts

(Henry, Shimoda, and Zhu, 2022), this finding has raised concerns that credit card may generate regressive transfers from naive lower-income consumers to their more financially sophisticated, higher-income counterparts (Felt et al., 2023; Agarwal et al., 2023; Egan et al., 2026).⁴

Nevertheless, subsequent empirical facts demonstrate that *Revolvers* actively utilize credit card debt to finance current consumption. Because this liquidity benefit is entirely omitted from net rewards, focusing exclusively on inequality in this metrics can lead to distorted and potentially misleading conclusions regarding the overall welfare effects of credit cards.

Fact 2: *Revolvers* Use Credit Card To Manage Liquidity

Fact 2 consists of several smaller facts showing that *Revolvers* use credit cards as a source of unsecured credit to finance consumption, and that these borrowing decisions reflect deliberate choices rather than mistakes.

These results suggest that, because net rewards do not capture the value of credit cards as a source of liquidity, this metric may severely underestimate the value of credit cards for *Revolvers*. Consequently, relying on net rewards alone may overstate the true effect of credit cards on inequality, especially since low-income consumers are overrepresented among revolving cardholders. Moreover, the finding that most revolving behavior, as captured by the DWPT, appears planned rather than accidental motivates the use of a rational framework for modeling consumer demand for credit card debt.

Fact 2a: *Revolvers* Use Revolving Debt To Finance Consumption

Because the survey does not directly report total consumption, I construct a measure of monthly expenditure by scaling each respondent’s reported personal credit card spending by their estimated credit card spending share.⁵ Throughout the empirical analysis, I use this expenditure as a proxy for consumption. Although this proxy does not distinguish between specific consumption categories, the resulting estimates closely align with aggregate benchmark measures of average Canadian expenditure excluding housing

⁴Because the reward rates for points-based cards are calculated using the unweighted averages of point-earning rates and point valuations, this variable is undoubtedly measured with error. Nonetheless, this measurement issue is unlikely to alter the core qualitative result. The average annualized interest rate on outstanding balances is an order of magnitude larger than the average reward rate.

⁵See Appendix 10.1 for full details on construction

costs (**statcan**).

Figure 3a plots the average expenditure of *Convenience users* and *Revolvers* by income quintile. Data show clear evidence of revolving cardholders having higher spending level relative to their non-revolving counterparts. On average, *Revolvers* spend about CAD 2,700 per month, compared with roughly CAD 2,300 among *Convenience users*. This gap persists after controlling for demographic and household characteristics, including income, age, household size, and number of children (Appendix 10.2).

To verify whether the additional expenditure among *Revolvers* is funded by credit card debt, I estimate the following model:

$$\log(\text{Expenditure}) = \beta_0 + \beta_1 \log(\text{Balance}) + \beta_2 \log(\text{Income}) + \beta_3 r \quad (2)$$

where *Balance* denotes respondents' self-reported outstanding revolving balance, and *r* represents the advertised annual interest rate.

The key parameter of interest is β_1 , which captures the elasticity of expenditure with respect to outstanding revolving balances. Although this should by no means be interpreted as causal evidence, positive comovement between expenditure and credit card debt would suggest that *Revolvers* use credit cards to finance spending, thereby allowing them to sustain higher expenditure levels. Since $\log(\text{Revolving Balance})$ is undefined for consumers with no outstanding balances, namely *Convenience users*, I estimate Eq.(2) only on the *Revolvers* subsample.

Figure 3b depicts the estimated relationship between consumption and outstanding balances. Data indicate a statistically and economically significant relationship between spending and revolving debt among revolving cardholders. In particular, a 1% increase in outstanding balances is associated with a 0.11% increase in expenditure level.

Fact 2b: Revolving And Non-revolving Cardholders Report Comparable Levels Of Financial Well-Being

If revolving debt were primarily driven by poor financial management (Heidhues and Kőszegi, 2010; Agarwal et al., 2023), we would expect *Revolvers* to hold fewer assets, carry greater liabilities, and report weaker overall financial well-being than their non-revolving counterparts. To assess whether this perception is supported by data, I compare several indicators of financial well-being across revolving and non-revolving cardholders.

I begin with balance-sheet measures. Figure 4 compares the distributions of asset holdings and financial liabilities between *Revolvers* and *Convenience*

users. Since these variables are reported in broad bins, I present their cumulative distributions to provide a clearer comparison across the two groups.

Data suggest that revolving and non-revolving cardholders report similar distributions of assets (Figure 4a) and liabilities (Figure 4b). Consistent with this visual evidence, Kolmogorov–Smirnov (KS) tests fail to reject the null hypothesis that *Convenience users* and *Revolvers* have the same asset (p -value = 0.6) and liability distributions (p -value = 0.4).

As prior work has found that *Revolvers* tend to face tighter liquidity constraints (Zhou, 2022; Dao, 2025), balance-sheet measures alone may not fully capture the financial strain experienced by these cardholders. To address this concern, I next examine respondents’ self-reported financial comfort as an additional measure of financial well-being.⁶ Figure 5 plots the cumulative distribution of financial comfort among revolving and non-revolving cardholders. The two distributions closely resemble each others: approximately 63% of *Convenience users* report a comfort level of 8 or higher, compared with 65% of *Revolvers*. This difference is statistically insignificant, as reflected by the KS test (p -value = 0.7). Moreover, controlling for the possibility that *Revolvers* and *Convenience users* interpret the financial-comfort scale differently does not materially affect the results (Dao, 2025).

Taken together, these facts suggest that revolving debt need not reflect financial mistakes alone; it may also arise from deliberate liquidity-management (Telyukova, 2013) or self-control strategies (Dao, 2025).

Fact 3: *Revolvers* Exhibit Weaker Demand for Credit Cards.

For inequality in net reward to accurately reflect the redistribution effects of credit cards, a necessary condition is that preferences for rebates must be homogeneous. This section documents how consumers’ payment choices vary by revolving status and suggests that this variation is likely driven by heterogeneity in *Convenience users*’ and *Revolvers*’ preferences for rewards. These findings therefore highlight the need for an alternative approach that can account for such heterogeneity.

Fact 3a: *Revolvers* Favor Debit Cards Whereas *Convenience Users* Prefer Credit Cards.

Figure 6 illustrates how average payment patterns differ between *Convenience users* and *Revolvers* across the income distribution. Credit cards are

⁶Respondents were asked to select from the scale of 1 to 10 “How comfortable are you managing your current debt load ?”

the dominant payment method among *Convenience users*, accounting for 65% of total consumption for this group. By contrast, *Revolvers* allocate only 40% of their expenditure to credit cards. Roughly 65% of *Revolvers*' non-credit card spending (i.e., 40% total spending) is directed toward debit cards, suggesting that low credit card usage among this group is not due to innate preferences for physical notes (i.e., cash) over plastics.

The gaps in credit card usage between revolving and non-revolving cardholders also appear when looking at absolute spending. On average, *Revolvers* spend roughly CAD 1,000 per month on credit cards, about 50% less than their non-revolving counterparts, despite having higher total expenditure (Table 3). Since cardholders earn rebate by transacting with credit cards, observed differences in usage undoubtedly contribute to the gap in net rewards between *Convenience users* and *Revolvers* (Fact 1).

Aside from the borrowing function, the main features that distinguish credit cards from debit cards are the rewards programs and insurance coverage offered by the former. Thus, observed gaps in credit card usage between *Convenience users* and *Revolvers* are suggestive of significant heterogeneity in preferences for these features.⁷

Fact 3b: *Revolvers* Are Less Responsive to Rewards Than *Convenience Users*

To formally examine whether cardholders respond differently to credit card features by revolving status, I estimate the following regression model:

$$SoW_i(\%) = \alpha_0 + \mathbf{1}_{Rev}\alpha_1 + \mathbf{X}_i(\boldsymbol{\beta} + \mathbf{1}_{Rev}\boldsymbol{\delta}) + \mathbf{Z}_i\boldsymbol{\gamma} + \epsilon_i \quad (3)$$

where SoW_i denote credit card's spending a percentage of total consumption. $\mathbf{X}_i$ is vector of products characteristics including reward rate, as a percentage of spending, and the number of insurance coverages included. Finally, $\mathbf{Z}_i$ is vector of control variables consisting of annual fees, income and bank fixed effects. Since the indicator function $\mathbf{1}_{Rev} = 1$ for *Revolvers*, $\boldsymbol{\delta}$ allows for cardholders' responses to changes in card features to depend on revolving status. Figure 7 reports the main parameters of interests, namely $\boldsymbol{\beta}$ and $\boldsymbol{\delta}$. The full regression results are reported in the Appendix (table XX)

Inline with previous findings from the literature (Ching and Hayashi, 2010; Arango, Huynh, and Sabetti, 2011; Wang, 2023), I find cardholders to

⁷Since *Revolvers* use only about 40% of their available credit limits (Appendix), the lower demand for credit cards among this group is unlikely to reflect binding credit constraints.

be highly sensitive to rewards (Figure 7a). Increase in reward rate is associated with positive changes in credit card’s SoW regardless of the revolving status. Nevertheless, *Revolvers*’ responses to rewards are 2.5 percentage point, or equivalently 22%, smaller than the baseline of non-revolving cardholders. Similar patterns emerge under a double/debiased machine learning specification (Chernozhukov et al., 2018; Bach et al., 2024), suggesting that results are robust to assumptions about functional form of Eq.(3) (Appendix 10.2, Figure 18).

The lower responsiveness to rewards among *Revolvers* is also consistent with their tendency to hold cheaper cards with less generous rebate programs (Table 11). One concern is that this pattern may reflect supply-side restrictions rather than demand-side choices. However, since revolving and non-revolving cardholders report similar credit ratings and debt-to-income ratios (Dao, 2025), both groups should qualify for broadly similar products ex ante. This suggests that differences in reward rates and annual fees are more likely to reflect consumer self-selection than supply-side restrictions.

Overall, these results are consistent with Fact 3a, which suggests that lower credit card usage among *Revolvers* is not driven by a preference for physical cash over plastic, but rather by weaker preferences for features unique to credit cards.

Figure 2: Distribution of net reward by revolving status

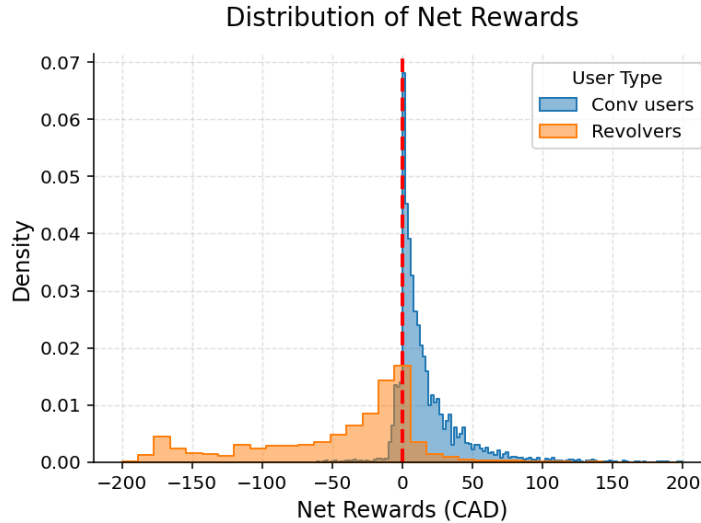
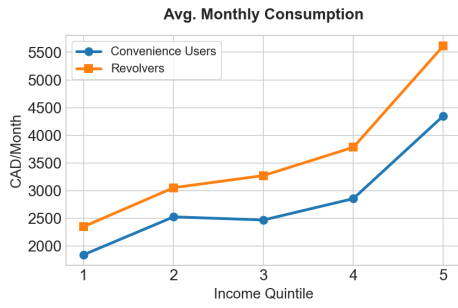


Figure 3: Consumption and Outstanding Balances

(a) Average consumption by income quintile



(b) Debt and consumption among *Revolvers*

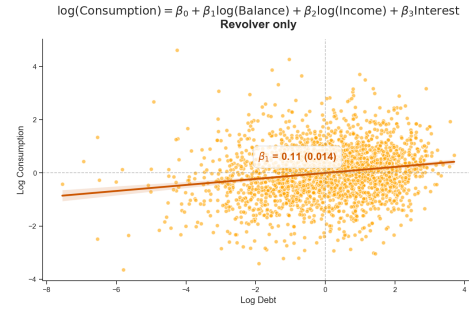
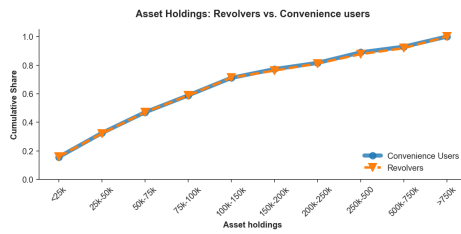


Figure 4: Cumulative Distributions Of Assets And Liabilities

(a) Assets holdings



(b) Financial Liabilities

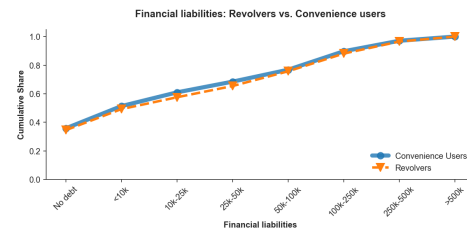


Table 3

Quintile	Credit Exp. (\$)		Cash Exp. (\$)		Debit Exp. (\$)	
	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.
1st (Bottom 20%)	876	579	287	386	481	1,050
2nd Quintile	1,171	827	338	568	676	1,198
3rd Quintile	1,452	962	301	411	504	1,353
4th Quintile	1,732	1,455	254	416	554	1,467
5th (Top 20%)	2,476	1,634	318	694	1,275	1,545

Notes: Conv. = Convenience users; Rev. = Revolvers.

Figure 5: Cumulative Distribution of Self-Reported Comfort

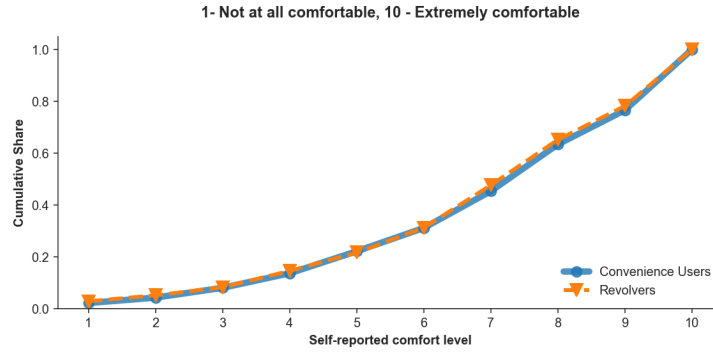


Figure 6: Variations in Payment Patterns

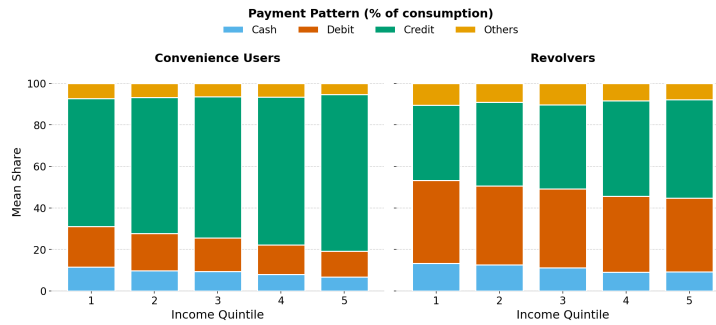
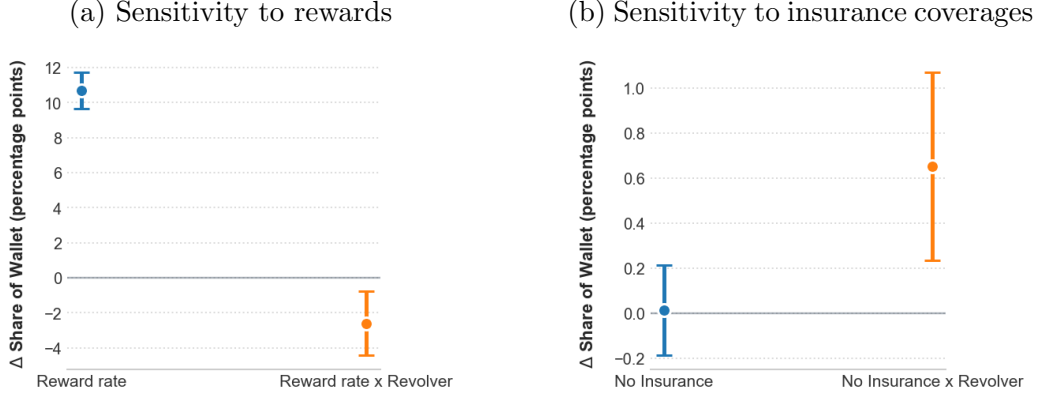


Figure 7: Heterogeneity In Consumer Preferences



4 Theoretical Framework

The literature has often interpreted revolving debt, and the resulting negative net rewards among *Revolvers* (Fact 1) as consequences of financial naivety (Heidhues and Kőszegi, 2010; Agarwal et al., 2023). Nevertheless, the evidence presented so far suggests that persistent revolving behavior, as captured by the DWPT, reflect deliberate attempt to manage liquidity rather than mistakes in financial planning (Fact 2).

This section proposes an alternative welfare measure based on consumer surplus from the introduction of credit cards (CS). I begin by introducing a partial-equilibrium model of credit card demand, taking the characteristics of adopted cards as given. The analysis therefore focuses on credit card usage conditional on card adoption and observed card characteristics, rather than modeling the upstream choice of which card to hold.

Because credit card borrowing links current consumption to future repayment obligations, modeling *Revolvers'* demand for credit cards requires a dynamic framework. This creates an identification challenge, as the DWPT provides only cross-sectional variation. Nevertheless, I show that consumer surplus from credit card use can be estimated under mild assumptions on the utility function.

The remainder of this section presents the general structure of the dynamic model and shows how consumer surplus can be calculated without estimating the full dynamic model. Formal treatments of the model are

relegated to Appendix 9.

4.1 A Simple Model Of Consumption and Payment Choice

Model Overview: Consider a standard finite-horizon consumption-savings problem in discrete time (Figure 8). At the beginning of each period t , consumers observe their state variables (y, d) , where y denotes income net of investment decisions and d denotes the current outstanding credit card balance. After observing (y, d) , consumers must make at least the minimum required payment on their credit cards θd where $0 < \theta < 1$. They then choose net credit card borrowing, Δd , and allocate consumption between cash-based methods, c , and credit cards, cc .

The model abstracts from additional savings within the consumption-payment stage. Thus, consumers must exhaust their available resources through consumption, implying the within-period budget constraint

$$y - \theta d + \Delta d = c + cc. \quad (4)$$

Outstanding balances evolve according the law of motion:

$$d' = (1 + r)((1 - \theta)d + \Delta d) \quad (5)$$

where r denotes per-period interest rate.

This setup captures the dual role of credit cards. The borrowing choice, Δd , reflects the demand of the credit function, while the allocation between c and cc reflects the use of credit cards as a payment instrument.

The model abstracts away from investment decisions and take y exogenously determined. This framing abstracts from the “credit card debt puzzle”, in which consumers simultaneously hold low-yield liquid assets and high-interest credit card debt (Bertaut, Haliassos, and Reiter, 2009; Telyukova, 2013). Instead, the model focuses on how consumers use credit cards to smooth consumption and manage payments, conditional on prior investment choices. Accordingly, readers should interpret y as disposable income net of investments and available for consumption. When consumers do not borrow through credit cards (i.e., $\Delta d = 0$), y coincides with their expenditure level.

Finally, as the credit function and the ability to earn rewards are unique to credit cards, I restrict the choice set to two payment methods: cash-based

methods and credit cards.⁸

Definition of revolving: In the context of this paper, net borrowing, denoted as Δd , reflects the net increase in the interest-bearing revolving balance during a given period. Letting R denote the amount repaid to the credit card issuer during that same period, Δd is defined as the difference between new credit card spending (cc) and repayment

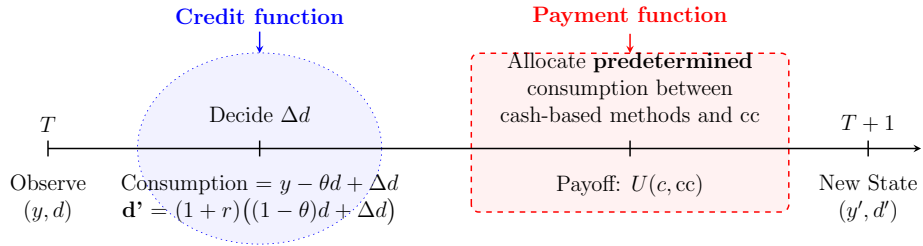
$$\Delta d = cc - R$$

I assume that *Convenience users* always repay their credit card balances in full and therefore do not carry any outstanding debt. For these consumers, $\Delta d = d = 0$ in every period. By contrast, *Revolvers* repay less than their credit card spending in at least some periods and therefore carry positive outstanding balances, so that $d > 0$.

Although the framework is flexible enough to accommodate both persistent and transitory revolving behavior, I focus on the former in the empirical implementation. The emphasis on long-term behavior is data-driven as the DWPT primarily captures persistent rather than transitory revolving behaviors. Nevertheless, the empirical analysis allows for general life-cycle dynamics in outstanding balances among *Revolvers*.

This flexibility is important because existing studies suggest that revolving debt rises during early-to-mid life-cycle periods before declining as consumers approach retirement (Fulford and Schuh, 2024; Dao, 2025). The model captures these dynamics by allowing *Revolvers* to either finance consumption with revolving credit (i.e., $\Delta d > 0$) or to pay down their balances (i.e., $\Delta d < 0$).

Figure 8: Model Timeline



⁸While Buy-Now-Pay-Later services allow consumers to borrow while using debit cards, these services were uncommon during the sample period.

4.2 Allocation Decision Is Static, Conditional On Consumption

Eq (4) and (5) clarify why, conditional on the net borrowing choice, consumers' allocation of total expenditure between cash-based methods and credit cards is a static decision.

For any given value of Δd , the current-period consumption (i.e., $c + cc$) is also uniquely determined. Since borrowing limits are not binding for most consumers (Table 11), these individuals can allocate their budget, $y - \theta d + \Delta d$, between payment methods without restrictions.

To illustrate this point, suppose the optimal level of net borrowing is fixed at $\Delta d = 100$. For simplicity, assume that $\theta = 0$ and $y = 900$. The total resources available for consumption is:

$$\begin{aligned} y &= 900, \Delta d = 100 \\ \Rightarrow y + \Delta d &= 900 + 100 = 1000 = c + cc \end{aligned}$$

Given this budget, there are multiple ways to allocate consumption across payment methods while keeping the same net borrowing choice.

Under Allocation A, the agent allocates \$400 to credit card spending and the remaining \$600 to cash-based methods. Since $\Delta d = cc - R$, this allocation implies that the agent repays \$300 toward their credit card balance.

$$\begin{aligned} \textbf{Allocation A : } \Delta d &= cc - R = 400 - 300 \\ y + \Delta d &= c + cc = 600 + 400 \end{aligned}$$

Under Allocation B, the agent allocates \$700 to credit card spending and \$300 to cash-based methods. To satisfy the same net borrowing choice, the agent must make a repayment of \$600.

$$\begin{aligned} \textbf{Allocation B : } \Delta d &= cc - R = 700 - 600 \\ y + \Delta d &= c + cc = 300 + 700 \end{aligned}$$

Both allocations imply the same net borrowing choice, $\Delta d = 100$. Consequently, they lead to the same next-period balance and have the same dynamic implications.⁹ Formal proof this is presented in Appendix 9.

⁹The timing of the borrowing and allocating decisions is not essential for this argument. In fact, the formal model assumes borrowing and payment-allocation decisions are made simultaneously.

Because the allocation of expenditure between cash-based methods and credit cards is static conditional on consumption, credit card usage depends on card-specific features and cardholders' preferences for those features. Thus, cross-sectional variation in usage and card features is sufficient welfare assessment. Section 4.3 explains the empirical approach.

4.3 From Continuous To Discrete Choice

When estimating demand for payment methods, a natural way to condition on consumption is to use relative expenditure shares. For example, rather than describing Allocation A as \$400 in credit card spending and \$600 in cash-based spending, one can express it as allocating 40% of consumption to credit cards and 60% to cash-based methods. This share-based formulation connects the theoretical payment-allocation problem to the empirical demand specification.

Suppose per-period utility takes a CES form in cash-based and credit card expenditures. The share of cash-based methods and credit cards in total expenditure can be expressed as conditional choice probabilities in a discrete choice framework under two assumptions:

Assumption 1.

A. Exogenous consumption level: *Individual's total consumption is taken as given.*

B. Constant transaction value: *Each cardholder faces a sequence of homogeneous transactions of fixed value $\bar{k}$.*

These assumptions map the continuous payment-allocation problem into a discrete choice framework and have been used in other empirical contexts (Ellickson, Grieco, and Khvastunov, 2020; Li, 2023). Assumption 1A is a technical requirement. By treating consumption level as given, we abstract from intertemporal borrowing decisions and focus exclusively on the static allocation of payment methods.

Assumption 1B arises out of necessity due to the lack of transaction-level data. By imposing homogeneity across transactions, Assumption 1B equates the share of transactions settled payment method j with its share of wallet as reported by the DWPT. As an illustration, consider an individual i facing

T transactions with value k_t . The share of consumption spent via method j (i.e., expenditure of j) is defined as:

$$\text{Expenditure share} = T \frac{s_{cc} \mathbb{E}[k_t | j \text{ is chosen}]}{\sum_1^T k_t}$$

where s_j , $\mathbb{E}[k_t | j \text{ is chosen}]$ are the share and expected value of transactions settled via j , respectively. $\sum_1^T k_t$ represents the total transaction volume (i.e., consumption level). Under Assumption 1B, this expression reduces to:

$$\text{Expenditure share} = T \frac{s_j \mathbb{E}[k_t | j \text{ is chosen}]}{\sum_1^T k_t} = T \frac{s_j \bar{k}}{T \bar{k}} = s_j$$

Under this environment, consumers' allocation across payment methods can be represented as a sequence of transactions, each with value k . For each transaction, consumers choose a payment method, $j \in \mathbf{J} = \{c, cc\}$. Because all transactions have the same value, the transaction share of method j coincides with its expenditure share. Section 4.4 discusses the implications of Assumption 1B for estimates of consumer preferences.

Since this representation of the decision problem does not depend on the value of k , I normalize $k = 1$ and interpret the problem as one in which consumers discretely allocate each dollar of expenditure between cash-based methods and credit cards.

Set of Identifiable Preferences: This paper highlights the set of identifiable preferences when using the discrete choice framework to study demand for payment (Ching and Hayashi, 2010; Koulayev et al., 2016). Specifically, the discrete approach can identify preferences only for card features that shift spending across payment alternatives (i.e., “static” features). Preferences for “dynamic” characteristics such as interest rates and credit limits, remain unidentified because they directly affect consumption decisions, thereby violating Assumption 1A. The same restriction applies more broadly to studies that discretize continuous choices (Ellickson, Grieco, and Khvastunov, 2020; Li, 2023): characteristics that may violate analogous conditional-static assumptions should not be included in the discrete choice problem.

Appendix XX provides an example illustrating the consequences of violating this restriction.

4.4 Demand Estimation

Indirect utility: For each individual, indirect utility is specified as:

$$U_{ijt} = \delta_{ij} + \epsilon_{ijt},$$

where U_{ijt} denotes the indirect utility that individual i receives from choosing payment method $j \in \mathbf{J} = \{c, cc\}$ for the t -th dollar. The term δ_{ij} captures the deterministic component of utility, while ϵ_{ijt} represents an idiosyncratic taste shock.

For each dollar of expenditure, consumers choose the payment method that yields the highest utility:

$$j^* = \arg \max_{j \in \mathbf{J}} U_{ijt}.$$

Denote by s_{ij} individual i 's expenditure share for payment method j . Assuming that ϵ_{ijt} follows a Type-I extreme value distribution (McFadden, 1977), s_{ij} has the following closed-form expression:

$$s_{ij} = \frac{\exp(\delta_{ij})}{\sum_{\ell \in \mathbf{J}} \exp(\delta_{i\ell})}.$$

The deterministic utility depends on the features of the specific cards held by consumers, as well as cardholders' preferences for those features. Formally,

$$\delta_{ij} = \alpha_i \text{Cashback}_j + \beta_i \text{Points}_j + \gamma_i \text{Insurance}_j + \Xi_{ij} \quad (6)$$

where Cashback, Points denote the cashback rate and cashback-equivalent rate of point-based cards, respectively.¹⁰ Insurance is the number of insurance coverages included with the credit card. Finally, Ξ_{icc} captures characteristics that are unobserved by the econometrician.

As illustrated above, Assumption 1B bypasses the need to observe specific transaction characteristics, most notably transaction values. This simplification is not completely innocuous as payment choices could be highly context-dependent. For instance, in high-value transactions, consumers generally prefer credit cards to banknotes (i.e., cash) for convenience and security

¹⁰Since I consider only the primary card, consumers receive either cashback or points on their credit cards.

concerns; whereas the latter is preferred for small-value purchases (Huynh, 2019). This concern is mitigated by the product nesting which defines “cash-based methods” to include both banknotes and debit cards. Since debit cards share many of the convenience and security advantages of credit cards but do not typically offer rewards, this nesting reduces the extent to which transaction value mechanically drives the comparison between cash-based methods and credit cards. Given this product nesting, the primary margin of differentiation between cash-based methods and credit cards becomes the features exclusive to the latter, most importantly reward programs. Because transaction value determines the total dollar amount of rewards earned, but not the marginal utility of a given reward rate (i.e., α_i and β_i), violation of Assumption 1B should not materially affect the results of this analysis.

Estimating Equation: Following the literature (Berry, Levinsohn, and Pakes, 1993), I decompose Ξ_{icc} into a component that is common across individuals and an individual-specific component. That is,

$$\Xi_{ij} = \xi_j + \mathbf{1}_{Rev}\psi_j + \mathbf{1}_{Point}\Psi_j + \xi_{ij}. \quad (7)$$

Since cash-based methods do not offer any rebates or insurance coverages, I normalize $\delta_c = \xi_{ic}$ and estimate consumer preferences under the following specification (Berry, 1994):

$$\begin{aligned} \log \frac{s_{icc}}{s_{ic}} &= \delta_{icc} - \delta_{ic} \\ &= \xi_{cc} + \mathbf{1}_{Rev}\psi_{cc} + \mathbf{1}_{Point}\Psi_{cc} + \alpha_i \text{Cashback}_i + \beta_i \text{Point}_i + \gamma_i \text{Insurance}_i \\ &\quad + \tau \mathbf{Z}_i + \xi_{icc} - \xi_{ic} \end{aligned} \quad (8)$$

where Cashback_i , Point_i , Insurance_i are defined as before. The indicator $\mathbf{1}_{Point} = 1$ for consumers with a point-based credit cards. Consumer preferences are modeled as:^{11, 12}

$$\begin{aligned} \alpha_i &= \alpha + \sigma_1 \text{Income}_i + \mathbf{1}_{Rev}\alpha_{Rev} \\ \beta_i &= \beta + \sigma_2 \text{Income}_i + \mathbf{1}_{Rev}\beta_{Rev} \\ \gamma_i &= \gamma + \mathbf{1}_{Rev}\gamma_{Rev} \end{aligned}$$

¹¹Descriptive evidence suggests that preference for insurance is constant across income group. As such, I do not interact *Insurance* with income to obtain more efficient estimates.

¹²Since both income and revolving status are directly observable in the DWPT, heterogeneity in consumer preferences can be captured directly through interaction terms between relative characteristics and demographics.

Finally, the vector $\mathbf{Z}_i$ includes bank fixed effects, annual fees, and consumers' satisfaction with their primary card to mitigate potential endogeneity concerns due to selection (Ching and Hayashi, 2010). The estimated preference parameters remain robust across alternative specifications of Eq.(6) (Appendix XX).

5 Measuring The Value Of Credit Cards

5.1 Consumer Surplus

Following the discrete choice literature (Small and Rosen, 1981), I measure consumer surplus (CS) as the compensating variation required to make individuals equally well off in a counterfactual setting without credit cards.¹³ Because the available data are measured at the monthly level, I compute monthly consumer surplus. In particular,

$$CS_i \text{ (CAD/month)} : \frac{U_{\text{with cc}}^{i*} - U_{\text{without cc}}^{i*}}{\alpha_i} - \frac{\text{Annual fee}_i}{12} - \text{Borrowing Costs}_i \quad (9)$$

Here, $U_{\text{with cc}}^{i*}$ and $U_{\text{without cc}}^{i*}$ denote individual i 's utility levels with and without credit cards, respectively.

For a transaction of value k , a cashback rate of Cashback_i reduces the consumer's effective price by $k \cdot \text{Cashback}_i$. Thus, Cashback_i enters the payment-choice equation as a price-like variable, while α_i captures the marginal utility of income. Thus, I divide $U_{\text{with cc}}^{i*} - U_{\text{without cc}}^{i*}$ by this coefficient to convert the utility difference into dollar terms. Additionally, I divide annual fee by 12 to express this cost on a monthly basis.

Finally, Borrowing Costs_i captures the flow interest costs associated with a specific instance of debt-financed expenses (i.e., Δd). This measure should be distinguished from the total interest charges used in the calculation of net rewards. Because total interest charges are calculated from total outstanding balance, which is comprised of multiple instances of unpaid debt (i.e., Δd). As a result, using total interest charges instead of Borrowing Costs_i would incorrectly assign the costs of multiple instances of borrowing to current-

¹³Appendix 9 establishes the connection between this empirical measure and the underlying dynamic mode

period expenditure and overstate the relevant cost of credit. Section 5.3 expounds on the estimation procedures of Δd_i and Borrowing Costs.

For notational brevity, individual index i is suppressed for the remaining of the paper. Unless stated otherwise, all variables remain individual-specific.

Utility With Credit Cards: Consumers' utility with credit cards is calculated as,

$$U_{\text{with cc}}^* = \mathbf{v}_{\text{with cc}} \text{Expenditure}_{\text{with cc}} \quad (10)$$

where $\text{Expenditure}_{\text{with cc}}$ is the level of expenditure observed from the data.

Because consumers choose the payment method that yields the highest utility, per-unit surplus with credit cards is given by the expected maximum of U_{ct} and U_{cct} .

$$\mathbf{v}_{\text{with cc}} = \mathbb{E}[\max\{U_{ct}, U_{cct}\}] = \log \sum_{j \in \{c, cc\}} \exp(\delta_j) + \gamma$$

with $\gamma \approx 0.577$ denotes the Euler-Mascheroni constant, emerging from the assumption that shocks are drawn from a Type-I Extreme Value distribution.

Utility Without Credit Cards: In a counterfactual without credit cards per-period utility is given by:

$$U_{\text{without cc}}^* = \mathbf{v}_{\text{without cc}} \text{Expenditure}_{\text{without cc}}$$

When credit cards are removed, consumers no longer choose the payment option that provides the highest utility between cash-based methods and credit cards. As a result, per-unit surplus without credit cards is:

$$\mathbf{v}_{\text{without cc}} = \mathbb{E}[U_{ct}] = \delta_c + \gamma$$

where $\text{Expenditure}_{\text{without cc}}$ denotes the counterfactual expenditure level without credit cards.

Since debit cards and cash do not provide a revolving-credit function, consumers who currently finance part of their expenditure through credit card borrowing would no longer be able to sustain that portion of spending. Hence, I calculate $\text{Expenditure}_{\text{without cc}}$ by reducing observed expenditure by the estimated share financed through new credit card borrowing.

$$\text{Expenditure}_{\text{without credit}} = (1 - s_d) \cdot \text{Expenditure}_{\text{with credit}} \quad (11)$$

where $s_d = \max\{0, \frac{\Delta d}{\text{Expenditure}_{\text{with credit}}}\}$ reflects the share of consumption funded by credit card debt.¹⁴

By construction, $s_d = 0$ for cardholders who do not increase their revolving balances, including *Convenience users* and *Revolvers* who are actively paying down existing debt. Replacing Eq.(11) into the formula of utility without credit cards gives:

$$U_{\text{without cc}}^* = \mathbf{v}_{\text{without cc}} \cdot (1 - s_d) \cdot \text{Expenditure}_{\text{with credit}} \quad (12)$$

Measuring The Value of Credit Cards: Substituting Eq (10) and Eq (12) into the expression for consumer surplus gives:¹⁵

$$CS = \frac{(\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}) \cdot \text{Expenditure}}{\alpha} + \frac{\mathbf{v}_{\text{without cc}} \cdot s_d \cdot \text{Expenditure}}{\alpha} - \frac{\text{Annual fee}}{12} - \text{Borrowing Costs} \quad (13)$$

where Expenditure_i is the observed spending level.

With respect to the dynamic model developed in Section 4.1, Eq.(13) is best interpreted as an approximation to the difference in surplus between the observed equilibrium and a counterfactual in which consumers retain their inherited credit-card balances but cannot use credit cards for current period. Appendix 9 explores the connection between the static and dynamic framework in more details.

Baseline Utility Without Credit Cards: In most empirical applications of the discrete choice framework, the utility of the baseline option is typically normalized to zero. Under this convention, the per-unit surplus captures only the relative utility gain from introducing credit cards rather than an absolute measure of welfare. In contrast, my analysis interprets $\mathbf{v}_{\text{with cc}}$ and $\mathbf{v}_{\text{without cc}}$ not as relative differences, but as absolute utility levels with and without credit cards. This departure from standard practice is necessary to properly assess the welfare impacts of credit cards. In particular, because Eq.(13) contains the product between debt-financed expenditure and the absolute per-unit surplus without credit cards, imposing $\delta_c = 0$ would artificially deflate the estimated value of credit cards for *Revolvers*.¹⁶

¹⁴Recall $\Delta d = cc - R$ is the differences between credit card spending and repayment.

¹⁵Detailed derivation of Eq .(13) is provided in [Appendix](#)

¹⁶Recall $\mathbf{v}_{\text{without cc}} = \mathbb{E}[U_{ct}] = \delta_c + \gamma$.

Because payment-choice data identify utility only up to a common additive constant, the absolute utility from cash-based consumption is not point identified (Train, 2009). For this reason, I calibrate δ_c as:

$$\delta_c = \max\{0, \Xi_c - \Xi_{cc}\} \quad (14)$$

The term $\Xi_c - \Xi_{cc}$ captures the degree to which an individual prefers the unobserved components of cash-based methods over those of credit cards.¹⁷ If a consumer derives strictly greater unobserved utility from cash than from credit (i.e., $\Xi_c - \Xi_{cc} > 0$), their absolute baseline utility must be at least as large as this premium.

This calibration of δ_c implicitly normalizes the utility of forgoing consumption to zero. Because individuals in the data chose to consume, revealed preference dictates that any chosen option $j \in \{c, cc\}$ must generate weakly nonnegative surplus. By bounding δ_{ic} at zero, this calibration provides a conservative lower bound for $\mathbf{v}_{\text{without cc}}$. Consequently, the resulting estimates of the credit function’s welfare impact should be interpreted as a lower bound. **Appendix YY** presents robustness checks using the standard assumption of $\delta_c = 0$. While such normalization mechanically reduces the estimated value of the credit function for *Revolvers*, the overall effect of credit cards on inequality remains robust.

Importantly, the specific calibration of δ_c has no impact on the improvement in per-unit surplus caused by the introduction of credit cards. Differences in per-unit surplus with and without credit cards can be calculated using the standard log-sum formula:

$$\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}} = \log(1 + \exp(\delta_{cc}))$$

5.2 Welfare Effects Of Credit Cards

The Payment Function: The payment function improves welfare by enhancing baseline consumption utility through increases in per-unit surplus ($\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}$). This enhancement operates through two distinct channels: a variety effect and a quality effect. The variety effect emerges as holding a credit card allows consumers to consistently choose the optimal payment alternative.¹⁸ Meanwhile, the quality effect operates through $\mathbf{v}_{\text{with cc}}$, which

¹⁷In practice, $\Xi_{ic} - \Xi_{icc} = -(\xi_{cc} + \mathbf{1}_{Rev}\psi_{cc} + \mathbf{1}_{Point}\Psi_{cc} + \beta_i\text{Point}_i)$

¹⁸Mathematically, this ensures that the expected value with a credit card is at least as high as it is without one (i.e., $\mathbf{v}_{\text{with cc}} = \mathbb{E}[\max\{U_{ct}, U_{cct}\}] \geq \mathbf{v}_{\text{without cc}} = \mathbb{E}[U_{ct}]$)

is an increasing function of the credit card’s deterministic utility (i.e., δ_{cc}). Because δ_{cc} depends on both the card’s specific features (e.g., reward programs) and the individual’s preferences for those features, Eq.(13) directly incorporates preference heterogeneity into the welfare assessment.

The Credit Function: The credit function provides access to liquidity, allowing consumers to sustain the debt-financed share of their observed expenditure. Because this channel increases consumption without directly changing per-unit surplus, its direct gross welfare contribution is measured by valuing debt-financed expenditure at the counterfactual per-unit surplus without credit cards: $\mathbf{v}_{\text{without cc}} \cdot s_d \cdot \text{Expenditure}$.

This function also affects welfare indirectly through its interaction with the payment function. Access to revolving credit expands the expenditure base over which cardholders derive benefits from the payment function. This interaction becomes apparent by decomposing the value of the payment function as

$$\begin{aligned} (\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}) \cdot \text{Expenditure} &= (\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}) \cdot (1 - s_d) \cdot \text{Expenditure} \\ &\quad + (\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}) \cdot s_d \cdot \text{Expenditure} \end{aligned}$$

The second term captures the payment-function surplus associated with debt-financed expenditure. By sustaining a higher level of consumption, the credit function allows *Revolvers* to derive additional benefits from credit cards as a payment method.

The Tug-of-War Between Payment And Credit: Although the payment and credit functions both contribute positively to gross individual welfare, their distributional effects may operate in opposite directions. The net effect of credit cards on inequality is therefore an empirical question. This ambiguity suggests narrow focus on inequality in net rewards may provide an incomplete account of the distributional effects of credit cards, as they omit the potentially equalizing welfare gains from the credit function (Felt et al., 2023; Agarwal et al., 2023; Egan et al., 2026).

The payment function is likely to exacerbate consumption inequality, as higher-income cardholders generally hold credit cards with more generous features than their lower-income counterparts (Table 11). The magnitude of this effects, however, relies on the degree to which preferences are heterogeneous. Consider reward programs, for example: if low-income cardholders were to derive systematically higher marginal utility from rebates, such heterogeneity could partially offset disparities in earning rates and attenuate the regressive effects of reward programs.

Meanwhile, since higher-income households generally have access to cheaper alternative sources of credit (Appendix), the liquidity provided by revolving debt disproportionately benefits lower-income households. By allowing this group to consume beyond what their current incomes alone would permit, the credit function narrows the baseline inequality in consumption and partially offsets the regressive effect of the payment function. The magnitude of this countervailing force depends on consumers' repayment behavior, which determines their reliance on debt (i.e., s_d).

5.3 Estimating Value Of The Credit Function

Eq.(13) provides a semi-structural sufficient-statistics approach to measuring the welfare effects of credit cards without requiring estimation of the full dynamic structural model. Since its components are either directly observed (e.g., expenditure and annual fees) or estimable using cross-sectional variation (e.g., per-unit surplus, s_{id}), the framework substantially reduces the data and computational requirements of the welfare analysis.

Although the overarching qualitative insights regarding the inequality-reducing effects of the credit function remain robust, the precise quantitative estimates of the value of credit are sensitive to alternative calibrations. This is, admittedly, a notable limitation of the approach. Appendix XX assesses the sensitivity of the estimates to these alternative specifications.

Share Of Debt-Funded Expenditure: Of the required components, the per-unit surplus with and without credit cards can be directly computed using the estimated preference parameters and observed card features. By contrast, isolating the share of consumption financed by new debt (s_d) and its associated Borrowing Costs presents a nontrivial empirical challenge.

Before proceeding, it is useful to clarify the time dimension of the relevant empirical objects. Because the DWPT reports average spending by payment method around the time of the survey, the quantities used in this section are best interpreted as respondent-level short-term average monthly measures. Accordingly, s_d is best understood as the short-term average monthly share of expenditure financed by new borrowing around the survey window.

Define the individuals' average share of debt-funded spending as:

$$s_d = \max\{0, \mathbb{E}\left[\frac{\Delta d}{Expenditure}\right]\}$$

Given this definition, reliance on debt depends only on the average flow of

new borrowing over total expenditure and not the total stock of outstanding balances (i.e., d). Because the DWPT reports only the latter, there is a fundamental mismatch between the required variable and the observed data. Furthermore, given that roughly 45% of *Revolvers* carry outstanding balances exceeding their total monthly expenditure (Figure 1), it is not possible to proxy s_d with the observed debt-to-consumption ratio. I resolve this issue by using the model from Section 4.1 to impute the average level of net borrowing, $\overline{\Delta d}$.

Consider the average within-period budget constraint as specified by Eq.(4)

$$\bar{y} - \bar{\theta d} + \overline{\Delta d} = \bar{c} + \bar{cc} = Expenditure$$

where c , cc , and $Expenditure$ denote the observed individual-level average monthly cash-based spending, average monthly credit card spending, and average monthly total expenditure, respectively.¹⁹ Similarly, $\bar{y}$ and $\bar{\theta d}$ represent average disposable income net of investment decisions and the average minimum required payment.

Since *Convenience users* neither carry outstanding balances nor incur new revolving debt, their average expenditure level is exactly equal to their disposable income:

$$\bar{y}_{Conv} = Expenditure_{Conv}$$

Revolvers' expenditure level, on the other hand, is given by

$$\bar{y}_{Rev} - \bar{\theta d} + \overline{\Delta d} = Expenditure_{Rev}$$

To calculate the average monthly changes in balances for a revolving individual i , we must first recover the minimum required payment and disposable income. While $\bar{\theta d}$ is easily calculable using available data, estimating $\bar{y}$ is more complicated.

In particular, $\bar{y}$ is perfectly observable for *Convenience users*, but is unobserved for *Revolvers* whose spending includes both disposable income and new borrowing. To solve this, I predict *Revolvers'* average disposable income using a Gradient Boosting Regressor trained on the observed expenditures of *Convenience users*.²⁰

¹⁹Recall that the DWPT reports respondents' average spending across different payment methods.

²⁰Gory details regarding the implementation of Gradient Boosting Regressor as well as calculation of the minimum required payment are available in [Appendix xx](#).

For this prediction-based empirical strategy to be valid, the following assumption is required.

Assumption 2. *Convenience users and Revolvers have the same expected disposable income level, conditional on gross income and other demographics. That is,*

$$\mathbb{E}[\bar{y}_i | Z_i, \mathbf{1}_{Rev} = 0] = \mathbb{E}[\bar{y}_i | Z_i, \mathbf{1}_{Rev} = 1]$$

where Z_i is a high-dimensional vector of demographic variables.

Since $\bar{y}$ represents disposable income available for consumption, meaning $\bar{y} = \text{Income} - \text{Investment}$, Assumption 2 requires that expected level of net investment conditional on X_i is the same between revolving and non-revolving cardholders.^{21, 22} Although revolving cardholders tend to hold a larger share of their wealth in illiquid assets than their non-revolving counterparts (Dao, 2025), this difference in portfolio composition does not inherently violate Assumption 2.

The assumption strictly requires that conditional on income and demographics, the total investment level remains equivalent across revolving status but places no restriction on the composition of those investments. Empirically, Assumption 2 finds support in the strikingly similar assets and liabilities profiles of *Revolvers* versus *Convenience users* (Fact 2b; Figure 4).

Substituting the predicted income into the *Revolvers*' expenditure equation yields our estimates of average net borrowing:

$$\widehat{\Delta d} = \text{Expenditure} - (\hat{y} - \theta d) \quad (15)$$

Given estimates of Δd , the average share of debt-financed expenditure is calculated as:

$$\hat{s}_d = \min\{SoW, \max\{0, \frac{\widehat{\Delta d}}{\text{Expenditure}}\}\}$$

where SoW denotes credit card usage as a percentage of total spending.

Since some revolving users could be actively paying off their balances ($\Delta d < 0$), the max operator captures the fact that these consumers no longer

²¹Here, “Investment” serves as a catch-all term for all non-consumption allocations. This encompasses both asset accumulation and debt-servicing outflows.

²²Because the DWPT reports pre-tax rather than disposable income, I estimate after-tax income for all individuals based on their reported gross income and province of residence. Details of this calculation is discussed in [Appendix](#).

rely on credit card debt to finance their expenditure. Furthermore, because cardholders must first use credit cards in order to borrow, the outer min operator ensures that $\hat{s}_d$ does not exceed total card usage.²³

Because $\hat{s}_d$ is constructed using predicted disposable resources, it is also subject to prediction error. The result is, nevertheless, robust under a robust under a steady-state approximation.²⁴

Borrowing Costs: The cost of credit is calculated as the total interest charges generated by a particular instance of Δd . Suppose an instance of Δd remains outstanding for $\tilde{T}$ periods, the total borrowing costs is calculated as

$$\text{Borrowing costs} = \Delta d - P(\tilde{T} + 1) \quad (16)$$

where $P = \theta d + R$ denotes the fixed stream of payment.

Calculation of $\tilde{T}$ is based on the assumption that all of the per-period stream of repayment is attributed to Δd . Such an assumption aligns with regulatory requirements that mandate lenders to credit payments toward the outstanding balance components carrying the highest interest rates (Government of Canada, 2026). Because all cardholders begin with a zero outstanding balance, the only balance subject to repayment is newly accumulated borrowing, namely Δd . Appendix 9.1 discusses how $\tilde{T}$ is calculated in more details and establishes the relationship between Eq.(16) and the underlying dynamic model.

An important qualification is that when total repayment is small, namely $\theta d + R \ll \Delta d$, the outstanding balance may contain multiple borrowing increments accumulated across periods. Because regulations do not specify how repayments should be allocated in such instances, I assume banks adopt a first-in, first-out allocation rule, where repayments are attributed to the oldest Δd . This assumption yields the same expression for Borrowing costs as with Eq.(16).²⁵

²³Such inconsistencies may emerge due to prediction and measurement errors in $\hat{y}$ and θd .

²⁴The steady-state approximation assumes $\frac{d'}{(1+r)} = (1 - \theta)d + \Delta d = d \Rightarrow \Delta d = \theta d$.

²⁵Because total interest charges depend only the aggregate outstanding balance, banks have no incentive to structure repayments in a way that artificially prolongs the revolving spell of Δd_t .

6 Results

6.1 Preferences For Credit Card Rewards Are Highly Heterogeneous

This section documents substantial heterogeneity in consumers’ preferences for credit card rewards. Table 5 reports the preference parameters estimated from Eq.(8), while Figure 10 plots average preferences for cashback and points across income deciles. These estimates show that cardholders’ valuation of credit card reward vary systematically not only by their demographic profiles but also by the types of rewards collected. In particular, *Revolvers* value rewards less than *Convenience users*, and preferences for cashback decline with income, whereas preferences for points remain comparatively stable.

These differences have important economic implications. The same advertised reward rate need not generate the same consumer surplus across cardholders. Thus, inequality in net rewards may not be reflective of the true impacts of credit cards on inequality.

Within income groups: Conditional on income, *Revolvers* place less value on credit card rewards than *Convenience users* (Table 5). This difference persists after controlling for income and the type of reward earned (Figure 10). These estimates are consistent with the earlier reduced-form evidence: revolving cardholders are less responsive to changes in reward earn rates than their non-revolving counterparts (Figure 7).

One possible explanation for the observed pattern is the lumpiness of reward payouts. Specifically, most issuers distribute rebates only once a year or after a certain amount of rebates has been accumulated.²⁶ Because *Revolvers* are more likely to be liquidity constrained (Zhou, 2022; Dao, 2025), they may place less value on rewards that are delayed or paid in discrete amounts.

Between income groups: Conditional on revolving status, preferences for cashback decline with income, whereas preferences for points remain relatively stable.²⁷ This divergence may reflect differences in the valuation of the two reward structures.

²⁶For example, BMO’s cardholder terms and conditions state that cashback rewards may be redeemed only once annually, in December. See BMO.

²⁷Although the decile-specific estimates of cashback preferences are imprecisely estimated, the interaction between cashback and income is statistically significant (Figure 5).

Because cashback programs provide rebates equal to a fixed percentage of the transaction value, they function as direct price discounts. For example, a 2% cashback rate on a \$100 purchase provides a \$2 rebate and is functionally identical to a 2% discount on the same purchase. As the marginal utility of money diminishes with income, consumer preferences for cashback also declines as income rises.

Point-based reward programs, by contrast, do not have a fixed monetary value; instead the value per point varies across redemption categories. As higher-income consumers are better positioned to redeem points in relatively high-value categories, namely hotel stays and flights, reward points allows these cardholders to extract greater monetary value per point. This flexibility enables high-income cardholders to extract greater monetary value from each point.

To assess whether the results are consistent with this mechanism, I calculate cardholders' willingness to pay (WTP) for points using the standard coefficient-ratio approach:

$$WTP_{Point} = \frac{\beta_i}{\alpha_i}$$

where α_i, β_i denote the estimated marginal utility of cashback and points, respectively.²⁸ Figure 11 presents the average WTP across income deciles.

Results indicate that WTP_{Point} generally increases with income and rises sharply in the highest income decile. This pattern is consistent with the proposed mechanism underlying the divergence in preferences for cashback and points. However, because the data do not reveal how consumers redeem their points, the evidence remains suggestive rather than conclusive.

That the type of reward earned matters is not a new observation, as evidenced by the myriad guides showing consumers how to maximize the value of their reward points. Nevertheless, this dimension of preference heterogeneity has received limited attention in the literature.

6.2 Consumer Surplus

Reliance On Credit Card Debt: Figure 11 reports the average estimated share of expenditure financed by revolving debt (i.e., s_d) by income deciles.

²⁸Since cashback is measured in monetary term, α_i is equivalent to a price-coefficient and thus is interpreted as marginal utility of income.

Revolvers in the lowest income decile rely heavily on revolving debt, financing 14% of their total expenditure with revolving debt. This reliance steadily declines with income, falling to 7% among the highest-income *Revolvers*. The magnitude of s_d indicates that the credit function constitutes a critical component of the value revolving cardholders derive from credit cards. Because the net rewards metric does not capture this liquidity value, it substantially understates the true utility of credit cards for these users.

Furthermore, that reliance on debt declines with income suggests that by narrowing the expenditure gap between low- and high-income households, the credit function may actively reduce inequality. This dynamic is explored further in Section 7.

Welfare Gains From Credit Cards: I next compare the distribution of consumer surplus with the distribution of net rewards. Figure 13 depicts the distribution of consumer surplus, as defined by Eq.(13), and compares it with the distribution of net rewards.

Recall that *Revolvers* not only own less reward-focused credit cards than *Convenience users* (Table 11), but also place lower value on these rewards (Figure 9). As such, the formers gain from the introduction of credit cards a surplus of 0.63 cents per dollar of expenditure, roughly 44% lower than the 1.12 cents enjoyed by non-revolving cardholders (Figure 12, Table 5). Disparities in per-unit surplus are partially offset by the fact that *Revolvers* utilize the credit function to finance a higher expenditure level. Thus, the two groups derive comparable total monthly surplus from credit cards: CAD 17 for *Revolvers* and CAD 22.5 for *Convenience users*.

This finding contrasts sharply with the conclusion implied by the distribution of net rewards in the existing literature (Agarwal et al., 2023). Conditional on revolving status, over 86% of cardholders derive positive surplus from credit cards. By contrast, fewer than 15% of *Revolvers* receive positive net rewards, compared with 84.6% of *Convenience users* (Figure 13a).

The opposing conclusions implied by consumer surplus and net rewards arise because the latter omit the value of the credit function. The omission is particularly consequential in the lower half of the income distribution, where *Revolvers* derive greater surplus from credit cards than *Convenience users*, despite receiving considerably less surplus per unit of expenditure (Table 5). Because low-income *Revolvers* rely heavily on revolving debt to finance expenditure (Figure 11), the surplus generated by the credit function more than compensates for their lower surplus of the payment function.

6.3 Measuring Inequality

Extrapolated to the entire Canadian market, credit cards generate an annual surplus of CAD 9.12 billion, or roughly 9% of all social security benefits distributed by the government.^{29, 30} Given the scale of these gains, it is important to determine whether consumer surplus is broadly shared or concentrated among a small subset of cardholders and to identify the mechanisms generating any such concentration.

Inequality in Consumer Surplus: At almost CAD 50 per month, cardholders in the highest income decile derive roughly 6.5 times more surplus than those in the lowest income decile and 30% more consumer surplus than the next-highest group (Table 5). This concentration reflects their tendency to hold premium cards with more generous rewards and high ownership of point-based cards, whose flexible redemption options insulate them from the declining valuation of cashback (Table 11, Figure 9). Consequently, the distribution of consumer surplus is highly unequally as indicated by $Gini_{CS} = 0.56$.³¹

This figure is 41% lower than the Gini calculated for the distribution of net rewards (i.e., $Gini_{\text{Net Rewards}} = 0.79$).³² Thus, while welfare gains from credit cards are indeed unevenly distributed across cardholders, net reward severely overstate the true extent of such inequality. Comparison between $Gini_{CS}$ and $Gini_{\text{Net Rewards}}$ reinforces the importance of incorporating the credit function into welfare assessment, especially as low-income *Revolvers* rely heavily on the revolving to fund their spending (Table 11).

Mechanism: Figure 14 plots Lorenz curves for per-unit surplus generated by credit cards, baseline expenditure, and the resulting consumer surplus. For each curve, the horizontal axis plots the cumulative share of cardholders, ordered from the lowest to the highest value of the corresponding variable, while the vertical axis plots their cumulative share of that variable.

²⁹The annual surplus is extrapolated assuming an average monthly surplus of CAD 22, a card ownership rate of 83.4%, and a total population of 41.4 million (i.e., CAD 9.12 billion = $22 \times 0.83 \times 41,400,000 \times 12$).

³⁰In 2025, total transfers in social security benefits reached CAD 99.6 billion (Source: Statistics Canada)

³¹For comparison, the Gini coefficient for disposable income in Canada is approximately 0.30. See World Bank.

³²For perspective, a Gini coefficient of one corresponds to complete concentration of the relevant variable in a single individual. A coefficient of 0.79 therefore indicates an extremely unequal distribution.

A point (x, y) indicates that the bottom $x\%$ of cardholders account for $y\%$ of the variable’s aggregate value. The 45-degree line represents perfect equality, with larger deviations from this line indicating greater inequality.

Because low-income consumers receive higher marginal utility from rewards (Figure 9), per-unit surplus varies only modestly across cardholders. Hence, the resulting distribution is fairly even with $Gini_{\text{per-unit}} = 0.25$ (Figure 14a). Conversely, total expenditure is substantially more concentrated among high-income cardholders, yielding a moderately high level of inequality with $Gini_{\text{Expenditure}} = 0.45$ (Figure 14b).³³ Since total consumer surplus depends on both per-unit surplus and expenditure, inequality in expenditure amplifies the relatively modest differences in per-unit surplus. Consequently, surplus is heavily concentrated among high-income consumers resulting in a highly unequal distribution of welfare gains (Figure 14c).

Figure 14 underscores the mechanism through which welfare gains from credit cards are distributed. Because consumer surplus scales with expenditure, credit cards magnify baseline inequality in expenditure level by generating larger total gains for cardholders who spend more. The degree of inequality in consumer surplus, therefore, depends critically on the underlying distribution of expenditure.

Consistent with this mechanism, a Shapley decomposition of R^2 shows that variation in expenditure accounts for nearly 75% of the explained cross-sectional variation in consumer surplus, compared with only 20% for per-unit surplus (Table 6). In this sense, inequality in how welfare gains from credit cards are distributed appears to reflect baseline expenditure inequality more strongly than differences in per-unit surplus, which are more directly shaped by competition and market power. This finding qualifies concerns that stronger competition for cardholders may significantly exacerbate inequality (Egan et al., 2026).

Before moving on to the counterfactual analysis, it is necessary to distinguish inequality in consumer surplus from the incremental effects of credit cards on overall welfare inequality.³⁴ Although former is highly unequally distributed, this does not translate to significant effects on the distribution of welfare. Relative to the no-card counterfactual, credit cards increase the Gini coefficient of total welfare by approximately 2% under the baseline calibration

³³See the note accompanying Figure 1 in the World Bank blog post.

³⁴ $W_{\text{with cc}} = W_{\text{baseline}} + CS$. Hence, the incremental effects of credit cards on overall inequality is the difference between $Gini_{W_{\text{with cc}}}$ and $Gini_{W_{\text{baseline}}}$.

and by 13% under a conditional upper-bound specification. Appendix 10.3 provides the construction of baseline welfare and the corresponding sensitivity analysis.

Table 4: Estimates of preferences

Cashback is the cashback earn rate in percentage.

Point value is the cashback-equivalent earn rate of point.

$\mathbf{1}_{Revolver} = 1$ for *Revolvers*.

$\mathbf{1}_{Point} = 1$ if the primary card earns point-based rewards.

<i>Cashback</i> (%)	1.4*** (.077)
<i>Cashback</i> $\times$ <i>Income</i>	-.024*** (.0080)
<i>Point value</i>	.41*** (.059)
<i>Point value</i> $\times$ <i>Income</i>	.0039 (.0050)
No. Insurance	.0062 (.0060)
$\mathbf{1}_{Revolver} \times \textit{Cashback}$	-.36*** (.086)
$\mathbf{1}_{Revolver} \times \textit{Point Value}$	-.24*** (.055)
$\mathbf{1}_{Revolver} \times \textit{Insurance}$	.047*** (.012)
$\mathbf{1}_{Point}$	.50*** (.077)
$\mathbf{1}_{Revolver}$	-.98*** (.071)
Constant	-2.2*** (.16)

Figure 9: Preferences for credit card rewards

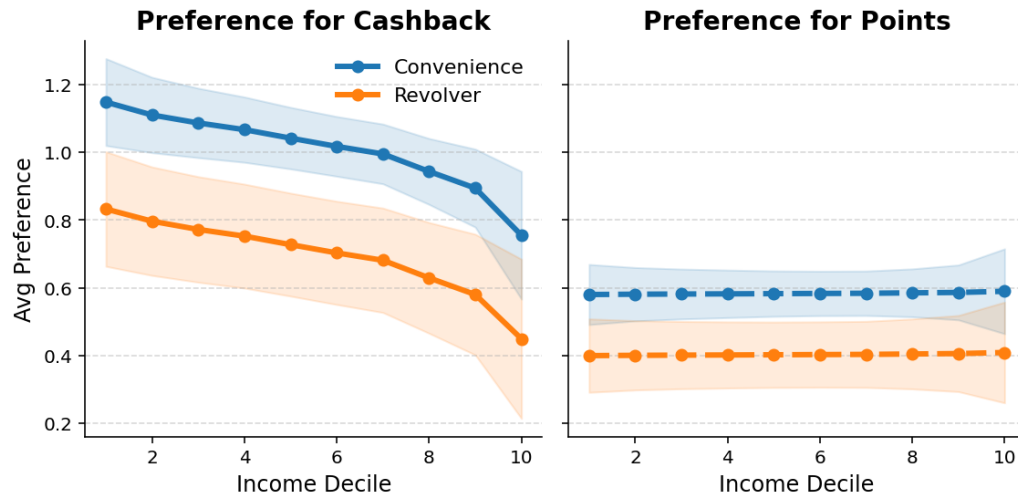
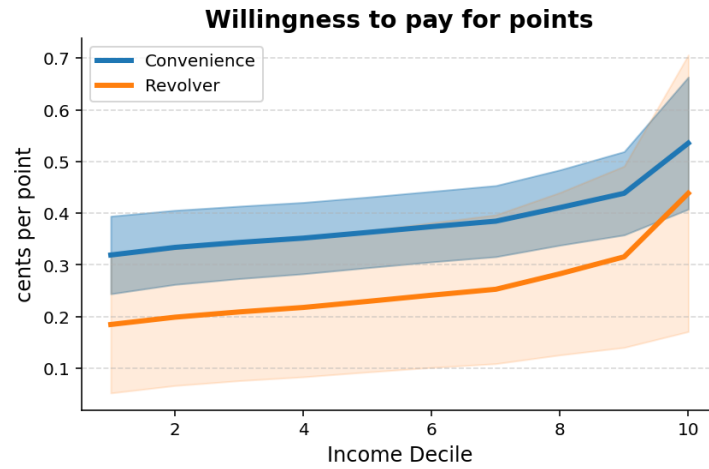


Figure 10: Average WTP for point



Standard errors were calculated following the delta method.

Figure 11: Average Share of Debt-Funded Expenditure

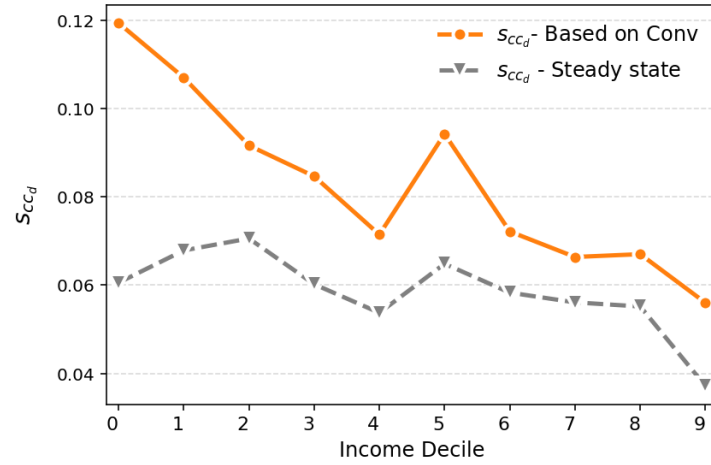


Figure 12: Per-unit surplus: *Convenience users* vs. *Revolvers*

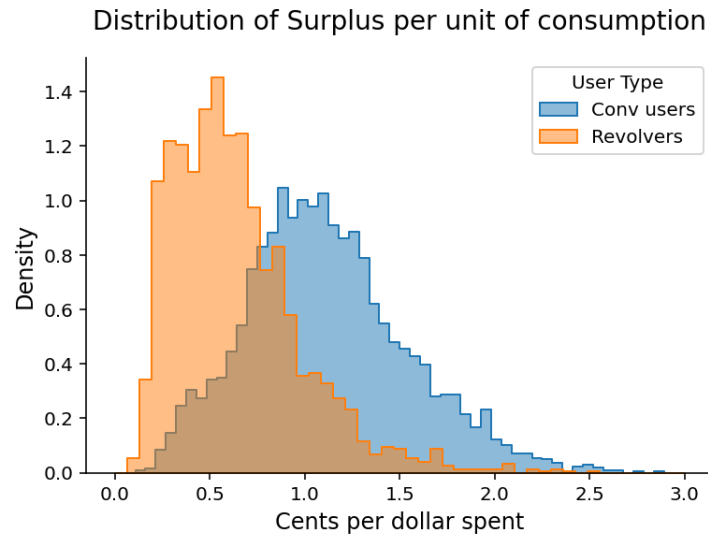


Figure 13: Distribution of Net Rewards and Consumer Surplus

Notes: Panel (a) reproduces the distribution of net rewards shown in Figure 2 to facilitate comparison. Panel (b) presents the corresponding distribution of consumer surplus. Both panels use the same sample and are expressed in monthly dollar terms.

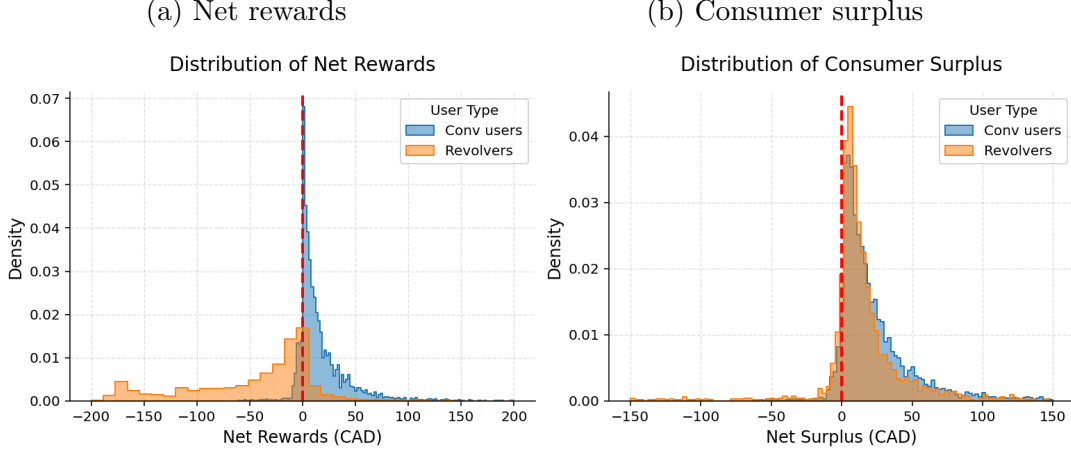


Table 5: Consumer Surplus and Net Rewards by Income Decile

Notes: Consumer surplus is winsorized at the 2.5th and 97.5th percentiles for the averages.

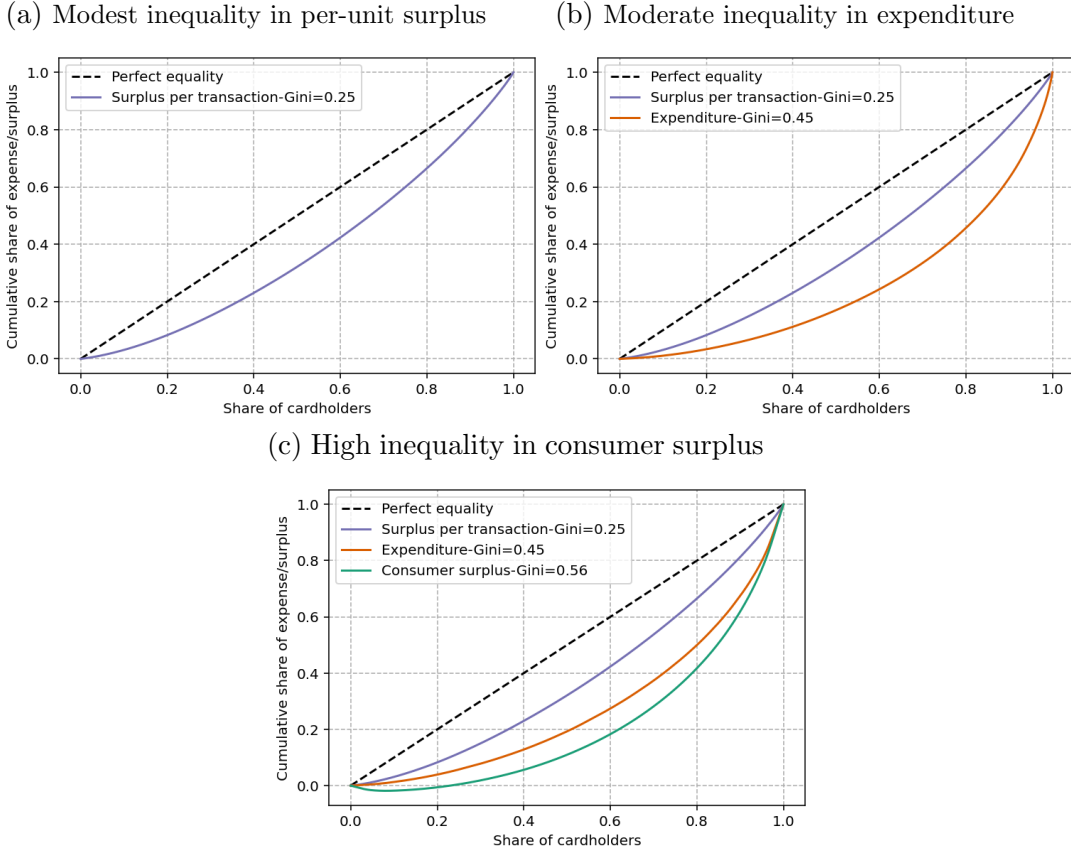
Metric	Type	Income Decile									
		1	2	3	4	5	6	7	8	9	10
Net rewards (CAD/month)	Conv	5.23	8.87	10.98	11.00	14.38	16.52	18.57	22.62	27.37	35.73
	Rev	-36.35	-47.79	-53.49	-44.79	-49.98	-46.55	-51.89	-49.93	-48.66	-36.13
	All	-6.81	-5.84	-5.68	-1.38	-0.35	4.62	4.14	8.41	14.32	25.52
Consumer surplus (CAD/month)	Conv	8.64	12.92	16.30	16.35	19.35	21.65	25.24	30.77	36.87	49.83
	Rev	4.14	9.73	10.46	12.12	17.13	14.00	24.82	28.15	36.00	48.47
	All	7.33	12.09	14.79	15.41	18.85	20.21	25.15	30.25	36.73	49.63
Per-unit surplus (cents/dollar)	Conv	0.73	0.88	0.94	0.98	1.05	1.10	1.21	1.29	1.51	1.78
	Rev	0.39	0.48	0.53	0.54	0.61	0.65	0.77	0.87	0.98	1.24
	All	0.64	0.78	0.84	0.88	0.95	1.02	1.12	1.20	1.42	1.70
Share of Revolvers		0.29	0.26	0.26	0.22	0.23	0.19	0.20	0.20	0.17	0.14

Table 6: Shapley Decomposition of Variation in Consumer Surplus

Notes: The table reports a Shapley decomposition of the R^2 from a regression of consumer surplus on total expenditure, per-unit surplus, borrowing costs, and annual fees. The final column reports each component's share of the total explained variation. The decomposition is implemented using the `shapley-decomposition` Python package (Itez, 2024).

Component	Shapley value	Share of explained variation
Total expenditure	0.513	62%
Per-unit surplus	0.150	18%
Borrowing costs	0.124	15%
Annual fees	0.038	5%
Total R^2	0.825	100%

Figure 14: Lorenz curves



7 Counterfactuals

As alluded to when discussing the welfare effects of credit cards, the payment and credit function can have countervailing effects on how the benefits of credit cards is distributed among cardholders. To quantify these effects, I simulate the distribution of welfare gains under two counterfactuals and compare each resulting Gini coefficient with the baseline $Gini_{CS}$.

The first counterfactual removes credit card rewards, thereby eliminating the main distributional channel associated with the payment function. Since reward programs are the main justification for annual fees (Table 10), I also set this fee to zero for the no-reward counterfactual. Consumer surplus under the no-reward counterfactual is given by:

$$CS_{\text{no rewards}} = (\mathbf{v}_{\text{no rewards}} - \mathbf{v}_{\text{without cc}}) \cdot \frac{\text{Expenditure}}{\alpha} + \mathbf{v}_{\text{without cc}} \cdot \frac{s_d \cdot \text{Expenditure}}{\alpha} - \text{Borrowing Costs} \quad (17)$$

where $\mathbf{v}_{\text{no rewards}}$ denotes the per-unit surplus when credit cards no longer give rewards.

The second counterfactual eliminates access to revolving credit while leaving reward programs intact. Since cardholders have to repay their balances in full at the end of the billing cycle, expenditure level is reduced by the amount previously financed through the credit function (i.e., s_d). This counterfactual is equivalent to removing the credit function of credit cards. Since cardholders can only utilize the payment function, consumer surplus under the no-borrowing counterfactual is calculated as:

$$CS_{\text{no borrowing}} = (\mathbf{v}_{\text{with cc}} - \mathbf{v}_{\text{without cc}}) \cdot \frac{(1 - s_d) \cdot \text{Expenditure}}{\alpha} - \frac{\text{Annual fee}}{12} \quad (18)$$

Figure ??, 16 and Table 7 summarizes the results of this exercise.

7.1 No-Rewards Counterfactual

Relative to the observed baseline, the Gini coefficient for consumer surplus declines by 12% in the no-rewards counterfactual, from 0.56 to 0.5. Reduction in inequality comes from a more equal distribution of per-unit surplus (Figure 15). Because *Revolvers* do not value rebates as much as *Convenience*

users, decreases in per-unit surplus are more severe for the latter than the former, thereby narrowing the disparity between the two groups. Another important equalizing force under the no-reward counterfactual is the elimination of annual fees. In particular, low-income *Revolvers* experience an increase in consumer surplus when reward programs are eliminated, as the reduction in annual fees more than offsets their loss of reward benefits (Table 7).

Reduction in inequality, unfortunately, comes at the cost of lower overall surplus. As rebate programs represent a major appeal of credit cards as a payment method (Ching and Hayashi, 2010; Wang, 2023), elimination of these program reduces total surplus level by over 25%, with the reduction being particularly pronounced among high-income *Convenience users* (Table 7). Average surplus among the highest-income non-revolving users declines by over 30%, as opposed to only 19% among revolving users in the same income bracket.

7.2 No-Borrowing Counterfactual

Conversely, eliminating access to revolving credit exacerbates inequality, increasing the Gini coefficient for consumer surplus by 6%, from 0.56 to 0.59 (Figure 16). Because only 21% of cardholders are *Revolvers*, restricting access to revolving credit directly affects a relatively small share of the cardholder population. As a result, the counterfactual has little effect on the distribution of total expenditure. This modest aggregate effect, however, masks substantial losses among *Revolvers*.

In particular, *Revolvers* experience sizable declines in consumption once the credit function is removed. These consumption losses translate into substantially lower consumer surplus, especially among middle-income *Revolvers*. Those in the fourth through seventh income deciles experience reductions in consumer surplus of over 30% under the no-borrowing counterfactual (Table 7). Given that *Revolvers* already derive less value from the payment function than *Convenience users*, these losses further widen the surplus gap between the two groups, thereby increasing inequality.

Figure 15: Lorenz curve under no reward counterfactual

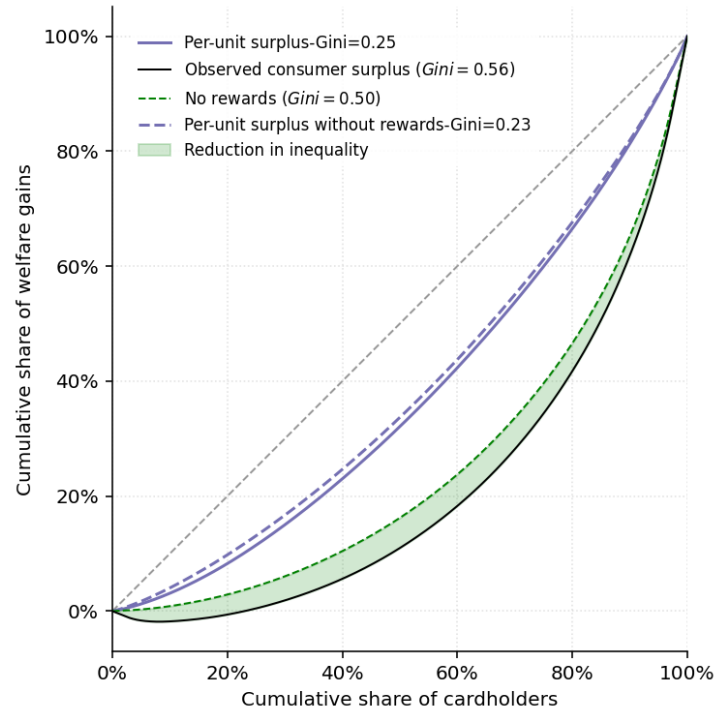


Figure 16: Lorenz curve under no borrowing counterfactual

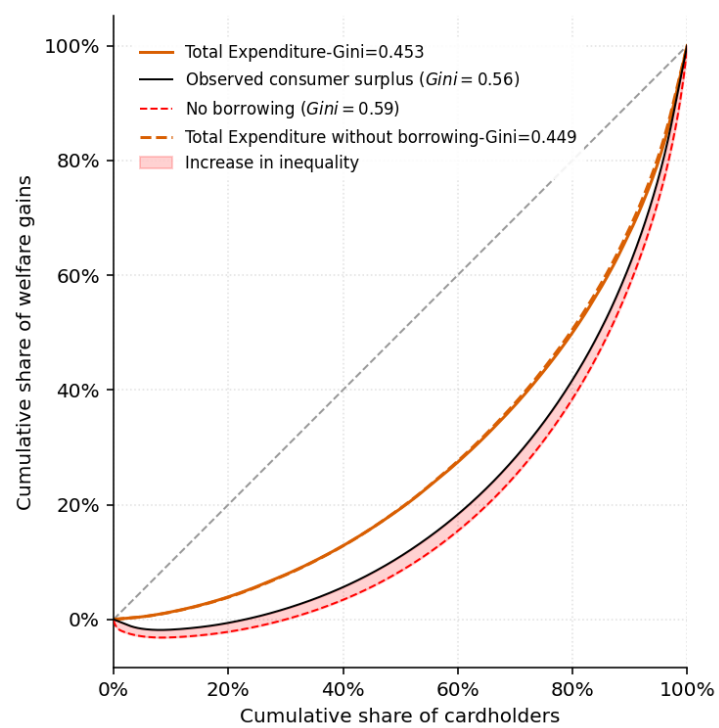


Table 7: Consumer Surplus Under Counterfactuals

Notes: Conv denotes *Convenience users*, and Rev denotes *Revolvers*.

Per-unit surplus is expressed in cents per dollar of expenditure, whereas Net rewards and Consumer surplus are in CAD per month.

All figures for consumer surplus are winsorized at the 2.5th and 97.5th percentile.

The no-rewards counterfactual removes credit card rewards and sets annual fees to zero while preserving access to revolving credit.

The no-revolving-credit counterfactual leaves rewards and annual fees unchanged but requires all cardholders to repay their balances in full at the end of each billing cycle.

Metric	Type	Income Decile									
		1	2	3	4	5	6	7	8	9	10
Consumer surplus											
No rewards – Level	Conv	6.46	9.38	11.81	12.00	13.96	14.95	17.68	21.67	26.12	34.84
	Rev	5.76	10.16	11.59	12.19	16.02	14.37	22.81	25.30	30.67	39.08
No rewards – Relative to baseline	Conv	-25.3%	-27.4%	-27.5%	-26.6%	-27.9%	-31.0%	-29.9%	-29.6%	-29.2%	-30.1%
	Rev	39.1%	4.4%	10.8%	0.6%	-6.5%	2.6%	-8.1%	-10.1%	-14.8%	-19.4%
No revolving credit – Level	Rev	4.85	7.48	7.17	8.40	11.48	8.79	17.31	20.47	26.55	41.33
No revolving credit – Relative to baseline	Rev	17.3%	-23.1%	-31.5%	-30.7%	-33.0%	-37.2%	-30.2%	-27.3%	-26.2%	-14.7%
Per-unit surplus											
No rewards – Level	Conv	0.44	0.52	0.57	0.57	0.61	0.63	0.71	0.75	0.89	1.05
	Rev	0.29	0.36	0.39	0.41	0.44	0.47	0.57	0.64	0.73	0.90
No rewards – Relative to baseline	Conv	-39.4%	-41.1%	-40.0%	-42.1%	-41.8%	-43.1%	-41.7%	-41.5%	-41.3%	-41.2%
	Rev	-26.5%	-25.8%	-25.9%	-24.9%	-27.6%	-28.2%	-25.8%	-26.1%	-25.5%	-27.8%

8 Conclusion

This paper revisits the distributional effects of credit cards by distinguishing between their roles as payment instruments and sources of unsecured credit. Existing studies typically evaluate these effects using net rewards, defined as rebates earned net of interest charges and fees. Although this measure captures the financial costs associated with revolving debt, it excludes the value cardholders derive from access to liquidity. As a result, net rewards provide an incomplete measure of welfare, particularly for *Revolvers*, whose demand for credit cards stem from both the payment function and the credit function.

To address this limitation, I measure the welfare gains from credit cards using consumer surplus, defined as the compensating variation required to hold utility constant in a counterfactual world without credit cards. The

empirical analysis combines payment behavior from the Bank of Canada’s Digital Wallet and Payment Trends Survey with detailed information on the rewards, fees, interest rates, and insurance benefits of respondents’ credit cards. This approach allows consumer surplus to incorporate both preference heterogeneity over card features and the value of expenditure financed through revolving credit.

The results show that credit cards generate economically meaningful welfare gains for most cardholders, including those who carry outstanding balances. Average monthly consumer surplus is approximately CAD 24 for *Convenience users* and CAD 17 for *Revolvers*. Extrapolated to the Canadian cardholder population, these estimates imply approximately CAD 9.1 billion in annual consumer surplus. These gains are nevertheless distributed unequally, resulting in a Gini coefficient of 0.56.

This inequality is considerable, but substantially smaller than that implied by net rewards. The Gini coefficient for net rewards is 0.79, approximately 41% higher than the corresponding measure based on consumer surplus. There are two reasons for this difference.

First, *Revolvers* place less value on reward features than *Convenience users*, so nominal differences in rebates do not translate one-for-one into differences in welfare. Second, net rewards deduct the costs of borrowing without accounting for the utility generated by the associated credit-financed expenditure. This omission is especially consequential for lower-income *Revolvers*, who finance a larger share of their expenditure through revolving debt.

The demand estimates also reveal an interesting pattern in preferences for different forms of credit card rewards. Consumers’ valuation of cashback declines with income, consistent with cashback functioning as a direct price discount. By contrast, preferences for points remain relatively stable across the income distribution. One possible explanation is that higher-income cardholders can obtain greater monetary value from a fixed number of points by using high-value, travel-related redemption options. This advantage may offset the declining marginal utility of money and help explain the relatively stable valuation of points across income groups.

The counterfactual analysis reveals an unexpected “tug-of-war” between the payment and credit functions. Eliminating rewards reduces the Gini coefficient of consumer surplus by 12%, confirming that reward programs are regressive. This reduction in inequality, however, comes at the cost significant reduction in aggregate surplus.

By contrast, eliminating access to revolving credit increases the Gini coefficient by 6%. The loss of liquidity disproportionately reduces surplus among lower-income *Revolvers*, indicating that the credit function partially offsets the regressive effects of rewards.

These findings qualify the conventional view that credit card borrowing is necessarily a source of regressive redistribution. Instead, they suggest that policies aimed at reducing borrowing costs may have progressive distributional effects.

Due to data limitation, the debt-financed share of expenditure must be inferred from differences in expenditure between revolving and non-revolving cardholders. Moreover, the welfare framework evaluates payment choices conditional on observed expenditure and is therefore less suited to counterfactuals that generate large behavioral changes in borrowing, card adoption, or consumption. Future work using panel data linking transactions, repayments, balances, and card characteristics could directly identify these dynamic responses and further distinguish deliberate consumption smoothing from borrowing driven by behavioral biases.

9 Theoretical Appendix

9.1 Calculation Of Borrowing Costs

Suppose consumers make a fixed payment P in each period toward the new borrowing Δd , beginning in period $t + 1$. The balance attributable to Δd immediately before the first payment is

$$b_0 = (1 + r)\Delta d.$$

After making the first payment P , the remaining balance is of Δd in the next period, b_1 , is ³⁵

$$b_1 = (1 + r)\Delta d - P.$$

Carrying this balance into period $t+2$, applying interest, and making another payment P gives

$$b_2 = (1 + r)^2\Delta d - P(1 + r) - P.$$

More generally, the remaining balance of Δd after j payments evolves according to

$$b_j = (1 + r)b_{j-1} - P,$$

and has the closed-form solution

$$b_j = (1 + r)^j\Delta d - P \sum_{s=0}^{j-1} (1 + r)^s.$$

Using the finite geometric sum,

$$b_j = (1 + r)^j\Delta d - P \frac{(1 + r)^j - 1}{r}.$$

Suppose the debt is fully repaid after $T + 1$ payments. Imposing the terminal condition

$$b_{T+1} = 0$$

gives

$$(1 + r)^{T+1}\Delta d = P \frac{(1 + r)^{T+1} - 1}{r}.$$

³⁵Note this the balance attributable to Δd which is not the same as total revolving balances d .

Equivalently,

$$P = \frac{r\Delta d}{1 - (1 + r)^{-(T+1)}}.$$

Solving instead for the repayment horizon yields

$$T + 1 = \frac{\log\left(\frac{P}{P - r\Delta d}\right)}{\log(1 + r)},$$

Because the consumer makes $T + 1$ payments of size P , the total payment obligation generated by Δd is

$$\text{Debt Service} = P(T + 1).$$

Total borrowing costs are defined as total debt service net of the original principal:

$$\begin{aligned} \text{Borrowing Costs} &= \text{Debt Service} - \Delta d \\ &= P(T + 1) - \Delta d. \end{aligned} \tag{19}$$

Substituting the expression for the repayment horizon gives

$$\text{Borrowing Costs} = P \frac{\log\left(\frac{P}{P - r\Delta d}\right)}{\log(1 + r)} - \Delta d. \tag{20}$$

The empirical implementation imposes:

$$P = \text{Minimum payment} + R$$

9.2 Dynamic Formulation

Environment: The economy consists of a unit mass of finitely lived agents who maximize their expected utility over T periods subject to a discounted factor- β . Time is discrete and indexed by $t = 0, 1, \dots, T$.

At the beginning of each period t , consumers observe their states (y, d) where y, d denote income net of investment decisions and current outstanding balance, respectively. Given (y, d) , agents allocate their income between cash consumption (c), credit card spending (cc), and additional repayment (R) subject to the budget constraint.

Following the literature (Fulford and Schuh, 2024), I assume that cardholders seek to avoid leaving negative wealth and therefore always aim to repay their debt (i.e., $d_T = 0$).³⁶ The Bellman equation of the problem is given by:

$$V(y, d) = \max_{c, cc} \{U(c, cc) + \beta \mathbb{E}[V(y', d') | cc, R]\} \quad (21)$$

subject to :

$$\text{Debt accumulation : } d' = (1 + r) \left((1 - \theta)d + cc - R \right)$$

$$\text{Budget constraint : } y - \theta d = R + c$$

$$\text{Income process : } y' = \bar{y} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\text{No investment : } d' \geq 0$$

$$\text{Credit limit : } d' \leq \bar{D} = k\bar{y}$$

Two features of the budget constraint warrant further discussions.

First, available income is reduced by θd . This restriction captures the mandatory minimum payment imposed by issuers. In reality, failing to meet the minimum required payment often results in account suspension forcing credit card spending to be zero. By invoking the Inada condition (i.e., $\lim_{x \rightarrow 0} U'(x) = \infty$), the model ensures that agents always optimally pay θd to maintain access to credit cards.

Second, credit card spending is not bound by the available liquid budget (i.e., $y - \theta d$). This lack of constraint enables cardholders to use credit cards to consume beyond their immediate income, thereby capturing the credit function.

From Continuous to Discrete: Since the income process is exogenous, the allocation of resources between cash consumption, credit card spending, and debt repayment affects future periods solely through the evolution of outstanding debt, d' . This observation motivates the following proposition

Proposition 1. *The dynamic problem established in Equation (21) can be decomposed into a two-stage decision process. First, consumers determine their optimal level of net borrowing, $\Delta d = cc - R$. Second, conditional on this borrowing decision, they allocate their consumption between cash-based methods and credit cards.*

³⁶Without such assumption cardholders would trivially borrow up to the following limit.

Proof. We first state the Bellman equation corresponding to Proposition 1. The consumer's problem can be written as:

$$\begin{aligned}
V(y, d) = \max_{\Delta d} & \left\{ U(c^*, cc^*) + \beta E[V(y', d')] \right\} \\
\text{s.t.} \quad & y' = \bar{y} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma^2) \\
& d' = (1 + r)((1 - \theta)d + \Delta d) \\
& 0 \leq d' \leq \bar{D}
\end{aligned} \tag{22}$$

where $(c^*(d'), cc^*(d'))$ are the solutions to:

$$\begin{aligned}
\max_c \quad & U(c, cc) \\
\text{s.t.} \quad & y - \theta d + \Delta d = cc + c
\end{aligned} \tag{23}$$

Since $d' = (1 + r)((1 - \theta)d + cc - R)$, any pair of (cc, R) such that $\tilde{c}c - \tilde{R} = cc - R = \Delta d$ should result in the same continuation value $E[V(y', d')]$. Thus, Eq.(21) can be written with Δd as the choice variable. That is,

$$V(y, d) = \max_{c, \Delta d} \left\{ U(c, cc) + \beta \mathbb{E}[V(y', d') | \Delta d] \right\} \tag{24}$$

subject to :

$$\begin{aligned}
\text{Debt accumulation : } & d' = (1 + r)((1 - \theta)d + \Delta d) \\
\text{Modified budget constraint : } & y - \theta d + \Delta d = cc + c \\
\text{Income process : } & y' = \bar{y} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma^2) \\
\text{No investment : } & d' \geq 0 \\
\text{Credit limit : } & d' \leq \bar{D} = k\bar{y}
\end{aligned}$$

Since y is exogenous and the law of motion for outstanding balances is identical in the two formulations, any given choice of Δd implies the same continuation value in Eq. 22 and Eq. 24. To establish equivalence between the two problems, it remains to show that they generate the same optimal within-period allocation, (c^*, cc^*) , for each feasible Δd . Once this is shown, the two formulations have the same objective function over Δd and therefore imply the same optimal borrowing choice, Δd^* .

To this end, notice that the FOC with respect to c , as defined by Eq.(24) is

$$\begin{aligned}\frac{\partial V}{\partial c} &= \frac{\partial U}{\partial c} + \frac{\partial cc}{\partial c} \frac{\partial U}{\partial cc} + \beta \mathbb{E} \frac{\partial V(y', d')}{\partial c} = 0 \\ &= \frac{\partial U}{\partial c} - \frac{\partial U}{\partial cc} + \beta \mathbb{E} \frac{\partial V(y', d')}{\partial c} = 0\end{aligned}$$

Since $\mathbb{E}[V(y', d')]$ does not depend on c , the FOC wrt to c reduces to:

$$\frac{\partial V}{\partial c} = \frac{\partial U}{\partial c} - \frac{\partial U}{\partial cc} = 0 \quad (25)$$

Assuming that the utility function exhibits diminishing marginal utility in c and cc (i.e., $U' > 0, U'' < 0$), the solution to Eq.(25) coincides with the optimal solution of the allocation problem defined in Eq.(23). $\square$

Having established equivalence between the original dynamic framework and the twp-stage framework, the next step is to illustrate how consumers' preference for credit card features can be recovered using a discrete choice framework.

Proposition 2. *Under CES preferences over c and cc . That is,*

$$U(c, cc) = U(CES(c, cc)) = U((v_1 c^\rho + v_2 cc^\rho)^{\frac{1}{\rho}})$$

Consumer demand for cash-based methods and credit cards can be estimated using the discrete-choice model implied by Assumption 1.

Proof. The FOCs of Eq.(23) with respect to c and cc are

$$\begin{aligned}\frac{\partial U}{\partial c} &= \frac{\partial U}{\partial CES} \frac{\partial CES}{\partial c} = \lambda, \\ \frac{\partial U}{\partial cc} &= \frac{\partial U}{\partial CES} \frac{\partial CES}{\partial cc} = \lambda.\end{aligned}$$

Therefore, at an interior optimum,

$$\begin{aligned}1 &= \frac{\partial U / \partial c}{\partial U / \partial cc} \\ &= \frac{\partial CES / \partial c}{\partial CES / \partial cc} \\ &= \frac{v_1 c^{\rho-1}}{v_2 cc^{\rho-1}}.\end{aligned}$$

Rearranging gives

$$\frac{cc}{c} = \left(\frac{v_2}{v_1}\right)^{\frac{1}{1-\rho}}.$$

Following the literature (Anderson and De Palma, 1992; Ellickson, Grieco, and Khvastunov, 2020; Li, 2023), I impose parametric assumptions on v_2 . That is,

$$\begin{aligned} v_1^{\frac{1}{1-\rho}} &= \exp(\sigma\delta_{ic}) \\ v_2^{\frac{1}{1-\rho}} &= \exp(\sigma\delta_{icc}) \\ \sigma &= \frac{\rho}{1-\rho} \end{aligned}$$

Taking log of $\frac{cc}{c}$ gives the same expression as in Eq.(8)

$$\log \frac{cc}{c} = \frac{1}{1-\rho}(\delta_{icc} - \delta_{ic})$$

□

9.3 Definition Of Surplus

Denote by $V_t(\cdot)$ and $H_t(\cdot)$ the value functions with access to credit cards and in the counterfactual without access to credit cards, respectively.

The value function with credit cards is given by:

$$\begin{aligned} V_t(y, d) &= \max_{\Delta d} \left\{ U(c^*, cc^*) + \beta E[V(y', d')] \right\} \\ \text{s.t. } y' &= \bar{y} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma^2) \\ d' &= (1+r)((1-\theta)d + \Delta d) \\ 0 &\leq d' \leq \bar{D} \end{aligned}$$

where c^*, cc^* are solutions to the allocation problem as defined by Eq.(23).

Similarly, the value function under the counter factually is:

$$\begin{aligned} H_t(y, d) &= \max_c \left\{ U(c^*) + \beta E[H(y', d')] \right\} \\ \text{s.t. } y' &= \bar{y} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma^2) \\ d' &= (1+r)(1-\theta)d \\ 0 &\leq d' \leq \bar{D} \end{aligned}$$

where $c^* = y - \theta d$.

Thus, the counterfactual corresponds to an environment in which consumers lose access to credit card but must still repay their existing outstanding balances. This differs from a counterfactual in which credit cards were never introduced. Such a scenario would imply an initial balance of $d = 0$. In the counterfactual considered here, however, the initial balance d is fixed at the observed level.

Under the debt-runoff counterfactual, the welfare gain from credit cards is measured as

$$V_t(y, d) - H_t(y, d). \quad (26)$$

By contrast, in a counterfactual in which credit cards were never introduced, the welfare gain would be measured as

$$V_t(y, d) - H_t(y, 0).$$

The latter comparison mechanically sets the outstanding balance to zero and therefore conflates the value of access to credit cards with the wealth effect of eliminating existing debt. While this distinction is moot for *Convenience users*, whose initial balance is zero, is consequential for *Revolvers*, who carry substantial outstanding balances.

9.4 Decomposition Of Consumer Surplus

If the consumer enters period t with balance d , repays a fraction θ , and takes new borrowing Δd , then next period's balance is

$$d' = (1 + r)[(1 - \theta)d + \Delta d].$$

Thus, the continuation value associated with new borrowing Δd is

$$\begin{aligned} V_{t+1}(y', d') - H_{t+1}(y', \tilde{d}) &= \left[V_{t+1}(y', d') - V_{t+1}(y', \tilde{d}) \right] \\ &\quad + \left[V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d}) \right]. \end{aligned} \quad (27)$$

Holding access to credit cards fixed, $V_{t+1}(y', d') - V_{t+1}(y', \tilde{d})$ isolates the marginal continuation-value effect of new credit card borrowing. Hence, this term captures the value coming from the credit function.

By contrast, $V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d})$ holds debt fixed at $\tilde{d}$, thereby ruling out new borrowing, and compares access to credit cards with reliance on cash-based payment methods only. This term therefore captures the value generated by the payment function of credit cards.

Payment Function: Consider the payment-function term:

$$V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d}).$$

Holding debt fixed at $\tilde{d}$, this term compares the value of expenditure when credit cards are available as a payment instrument with the value of expenditure when only cash-based methods are available.

If available expenditure in period $t + 1$ is

$$Expenditure_{t+1} = y' - \theta \tilde{d},$$

Then,

$$CES(c, cc) = (v_1 c^\rho + v_2 cc^\rho)^{\frac{1}{\rho}} = Expenditure \cdot \psi_{cc}$$

where $\psi_{cc} = \left(v_1^{\frac{1}{1-\rho}} + v_2^{\frac{1}{1-\rho}} \right)^{\frac{1-\rho}{\rho}}$

Similarly, without credit card the CES aggregator yields,

$$CES(c) = Expenditure \cdot \psi_0$$

where $\psi_0 = v_1^{1/\rho}$

To isolate the value generated by the payment function, I impose that future balances are identical across the with-credit-card and without-credit-card environments, so that $d'' = \tilde{d}''$. This restriction applies only to calculation of $V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d})$ and effectively shuts down differences arising from new borrowing. Formally,

$$\begin{aligned} V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d}) &= U(Expenditure_{t+1} \psi_{cc}) - U(Expenditure_{t+1} \psi_0) \\ &\quad + \beta \mathbb{E} \left[V_{t+2}(y'', \tilde{d}'') - H_{t+2}(y'', \tilde{d}'') \right] \end{aligned} \quad (28)$$

Suppose further that future payment-function gains are sufficiently small that the discounted continuation component is negligible. That is,

$$\sum_{\tau=t+2}^T \beta^{\tau-t-1} [U(Expenditure_\tau \psi_{cc}) - U(Expenditure_\tau \psi_0)] \approx 0.$$

Thus, the payment-function component can be approximated by its one-period flow value:

$$V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d}) \approx U(\text{Expenditure}_{t+1}\psi_{cc}) - U(\text{Expenditure}_{t+1}\psi_0).$$

Equivalently, this expression can be interpreted as the one-period flow benefit of the payment function.

Assuming the underlying utility function is homogeneous of degree 1,

$$U(\text{Expenditure}_{t+1}\psi_{cc}) - U(\text{Expenditure}_{t+1}\psi_0) = \text{Expenditure}_{t+1}(\psi_{cc} - \psi_0)U(1). \quad (29)$$

Thus under the continuous model, welfare gains from the payment function is identified up to a multiplicative constant. The formula for consumer surplus under a discrete choice framework requires that $U(1) = 1$.³⁷

Since $\frac{\text{Expenditure}_{t+1}(\psi_{cc} - \psi_0)}{\text{Expenditure}_{t+1}} = (\psi_{cc} - \psi_0)$, this term captures the per-unit surplus gains from the introduction of credit cards. Readers interested in how the log-sum formula can appear under this framing are directed to Dubé, Joo, and Kim, 2025.

Credit Function: In period t the welfare gains from incurring Δd is given by:

$$V_t^{\Delta d > 0}(y, d) - V_t^{\Delta d = 0}(y, d) = U_t(y - \theta d + \Delta d) - U_t(y - \theta d) + \beta \mathbb{E} \left[V_{t+1}(y', d') - V_{t+1}(y', \tilde{d}') \right]$$

Assuming a constant shadow price of income (i.e., $U'(c) = \alpha$), the period- $t + 1 + k$ flow-utility difference is

$$U(y_{t+1+k}, d_{t+1+k}) - U(y_{t+1+k}, \tilde{d}_{t+1+k}) = \alpha \theta (d_{t+1+k} - \tilde{d}_{t+1+k}).$$

the continuation-value difference is

$$\mathbb{E}_t \left[V_{t+1}(y_{t+1}, d_{t+1}) - V_{t+1}(y_{t+1}, \tilde{d}_{t+1}) \right] = \alpha \theta \sum_0^{\tilde{T}} \beta^k (d_{t+1+k} - \tilde{d}_{t+1+k})$$

where $\tilde{T}$ denotes the final period after which Δd is completely payoff.

To isolate the interest cost of borrowing, consider a counterfactual scenario involving an interest-free loan of the same size, Δd_t , with an identical

³⁷The most obvious choice $U(CES) = CES$.

principal-repayment schedule. The difference in the outstanding balance between the two scenarios is:

$$\theta(d_{t+1+k} - \tilde{d}_{t+1+k}) = rb_k$$

where b_k is the part of the remaining outstanding balance attributable to Δd as defined in Appendix 9.1.

Thus the difference in minimum required payment between these scenarios is generated by the outstanding component of Δd .

The discounted streams of differences in utility yields:

$$\mathbb{E}_t \left[V_{t+1}(y_{t+1}, d_{t+1}) - V_{t+1}(y_{t+1}, \tilde{d}_{t+1}) \right] = -\alpha r \sum_{k=0}^{\tilde{T}} b_k$$

Note the term $\sum_{k=0}^{\tilde{T}} b_k$ captures the total amount of interest charges generated by Δd . From Appendix 9.1, b_k evolves according to:

$$\begin{aligned} b_{k+1} &= (1+r)b_k - P \\ (1+r)b_k - b_{k+1} &= P \\ \Rightarrow \sum_{k=0}^{\tilde{T}} (1+r)b_k - b_{k+1} &= (\tilde{T}+1)P \\ \Rightarrow \sum_{k=0}^{\tilde{T}} rb_k + \sum_{k=0}^{\tilde{T}} (b_k - b_{k+1}) &= (\tilde{T}+1)P \end{aligned}$$

Given $b_0 = \Delta d$ and $b_{\tilde{T}} = 0$, $\sum_{k=0}^{\tilde{T}} (b_k - b_{k+1}) = b_0 - b_1 + b_1 - b_2, \dots - b_{\tilde{T}} = \Delta d$

Thus, the value of the credit function is given by:

$$\begin{aligned} V_t^{\Delta d > 0}(y, d) - V_t^{\Delta d = 0}(y, d) &= U_t(y - \theta d + \Delta d) - U_t(y - \theta d) + \beta \mathbb{E} \left[V_{t+1}(y', d') - V_{t+1}(y', \tilde{d}') \right] \\ &= \alpha \Delta d - \alpha \beta ((\tilde{T}+1)P - \Delta d). \end{aligned}$$

where the second term is the same as with Borrowing costs term in the static formulation when $\beta = 1$.

Total consumer surplus: Collecting term gives the final expression for consumer surplus under the dynamic model, which coincidences with the static formulation in Eq.(9)

$$\begin{aligned}
V_{t+1}(y', d') - H_{t+1}(y', \tilde{d}) &= \left[V_{t+1}(y', \tilde{d}) - H_{t+1}(y', \tilde{d}) \right] + \left[V_{t+1}(y', d') - V_{t+1}(y', \tilde{d}) \right] \\
&= \text{Expenditure}_{t+1}(\psi_{cc} - \psi_0) + \alpha \Delta d - \alpha r \Delta d_t \frac{1 - [\beta(1+r)(1-\theta)]^{\tilde{T}}}{1 - \beta(1+r)(1-\theta)}.
\end{aligned}$$

10 Appendix

10.1 Variable Construction

Earning Rate of Point-Based Cards: Since most issuers operate their own rewards programs, with different point-earning rates and redemption schemes, comparing point-based cards poses a non-trivial challenge. To address this issue, I convert all point-based products into a cashback-equivalent earning rate by multiplying the unweighted average point-earning rate by the unweighted average value per point. Consider the TD First Class Travel Visa Infinite as an example. For this card, consumers earn an average of 5 points per dollar (Table 8), with an average point value of 0.333 ¢ per point (11). The cashback-equivalent earning rate is therefore $5 \times 0.333 = 1.665$ ¢ per dollar spent, or equivalently 1.665%.

While intuitive, this approach abstracts from the fact that consumers may redeem points across different reward options. For example, low-income *Revolvers* would likely convert points into statement credits, whereas high-income *Convenience users* may derive greater value by redeeming points for travel-related rewards. Although this simplification may introduce measurement error, it is nevertheless necessary as the DWPT does not report how consumers redeem their points.

Consumption: Since the survey does not directly report respondents' total consumption or expenditure, I construct this measure by combining respondents' reported credit card spending for "personal purposes" with the share of total spending paid by credit card.³⁸ Because the DWPT does not define precisely what constitutes personal spending, this measure of consumption is admittedly coarse and does not distinguish between durable and non-durable consumption. As a result, the analysis should be interpreted as capturing overall consumption rather than spending on any specific category.

³⁸The exact wording of the question is: "In a typical month, how much do you spend on your credit card(s) for personal spending? Please enter a dollar amount."

Nonetheless, at roughly CAD 2,500 per month, the total expenditure measure aligns well with data on current consumption, defined as total consumption net of housing costs **statcan**.

Table 8: TD First Class Visa Travel Infinite Earn Rate

Type of Purchases	Points per dollar
Travel through Expedia [®]	8 pts/\$
Groceries	6 pts/\$
Recurring bills	4 pts/\$
Other purchases	2 pts/\$

Source: FinlyWealth Points Calculator

10.2 Double ML

Fact 1 establishes a stark difference in *Revolvers*’ and *Convenience users*’ payment patterns. On average, revolving cardholders have substantially lower credit card usage in comparison to their non-revolving counterparts, both in terms of transaction volume and share of total expenditure. This divergence is robust, persisting across all income deciles (Figure 1).

While disparity in reward earning rates between *Revolvers* and *Convenience users* explains part of observed differences in usage, data suggest that preferences also play a significant role. To provide causal evidence of this heterogeneity, I estimate the following partially linear model:

$$\text{CC Usage (\% of consumption)} = g(\mathbf{1}_{Rev}, \mathbf{X}) + \nu \quad (1.1)$$

$$m(\mathbf{X}) + \epsilon > \bar{Z} \Leftrightarrow \mathbf{1}_{Rev} = 1 \quad (1.2)$$

where $\mathbf{X}$ is a high-dimensional vector consisting of card attributes (e.g., reward earning rates, insurance coverage), demographic controls (e.g., income, marital status, household sizes), and their interactions. The nonparametric function $g(\mathbf{X})$ captures the baseline relationship between card usage and $\mathbf{X}$. Similarly, $m(\mathbf{X})$ characterizes non-random component of the borrowing decision. Finally, θ quantifies the difference in credit card usage between *Revolvers* and *Convenience users*, conditional on all observable characteristics (i.e., the “treatment effect”).

Following the Double/Debiased Machine Learning methodology, I treat m and g as nuisance functions and estimate them using the Random Forest algorithm.³⁹ To mitigate regularization bias stemming from the estimated $\hat{m}$ and $\hat{g}$ (Chernozhukov et al., 2018; Bach et al., 2024), we compute the individual-level orthogonal scores as:

$$\tilde{\theta}_{Revolve}^i = \hat{g}(1, \mathbf{X}_i) - \hat{g}(0, \mathbf{X}_i) + \frac{\mathbf{1}_{Rev}(Y_i - \hat{g}(1, \mathbf{X}_i))}{\hat{m}(\mathbf{X}_i)} - \frac{(1 - \mathbf{1}_{Rev})(Y_i - \hat{g}(0, \mathbf{X}_i))}{1 - \hat{m}(\mathbf{X}_i)}$$

Assuming $g(D, X) = g(X) + \text{Rewards}_i \cdot \mathbf{1}_{Rev} \cdot \delta_{Rev}$. The Conditional Average Treatment Effects (Semenova and Chernozhukov, 2020) is estimated as:

$$\tilde{\theta}_{Revolve}^i = g(1, X) - g(0, X) = \beta_0 + \delta_{Rev} \cdot \text{Rewards}_i + \xi_i \quad (2)$$

The main parameter of interest δ_{Rev} captures difference in responsiveness with respect to reward earning rate (i.e., Rewards_i) between *Revolvers* and *Convenience users*. Individual-level treatment effects, $\tilde{\theta}_{Revolve}^i$, are estimated using a doubly robust framework (Robins, Rotnitzky, and Zhao, 1994; Imbens and Wooldridge, 2009), which protects against misspecification bias in either the outcome model or the selection model, so long as one is correct.⁴⁰

To assess whether consumer preferences also depend on the specific reward type, I estimate the following analog of Eqs (1)-(2):

$$\text{CC Usage (\% of consumption)} = h(\mathbf{1}_{Point}, \mathbf{X}) + \mathbf{1}_{Point}\theta_{Point} + \tilde{\nu} \quad (3.1)$$

$$k(\mathbf{X}) + \tilde{\epsilon} > \bar{P} \Leftrightarrow \mathbf{1}_{Point} = 1 \quad (3.2)$$

where $h(\cdot), k(\cdot)$ are the equivalence of $g(\cdot), m(\cdot)$. The individual-level “treatment effects” (i.e., θ_{Point}) and CATE (i.e., δ_{Point}) are estimated following similar procedure as with $\theta_{Revolve}, \delta_{Revolve}$. That is,

$$\begin{aligned} \tilde{\theta}_{Point}^i &= \hat{h}(1, \mathbf{X}_i) - \hat{h}(0, \mathbf{X}_i) + \frac{\mathbf{1}_{Point}(Y_i - \hat{h}(1, \mathbf{X}_i))}{\hat{k}(\mathbf{X}_i)} - \frac{(1 - \mathbf{1}_{Point})(Y_i - \hat{h}(0, \mathbf{X}_i))}{1 - \hat{k}(\mathbf{X}_i)} \\ \tilde{\theta}_{Point}^i &= \beta_0 + \delta_{Point} \cdot \text{Rewards}_i + \tilde{\xi}_i \end{aligned} \quad (4)$$

³⁹`sklearn's RandomForestClassifier and RandomForestRegressor`

⁴⁰If the outcome model is correctly specified, $Y_i(1) - \mathbb{E}[\hat{g}|\mathbf{X}, 1] = Y_i(0) - \mathbb{E}[\hat{g}|\mathbf{X}, 0] = 0$ and $ATE = \mathbb{E}[\tilde{\theta}_{Revolve}^i] = \mathbb{E}[\hat{g}(1, \mathbf{X}_i) - \hat{g}(0, \mathbf{X}_i)] = g(1, \mathbf{X}_i) - g(0, \mathbf{X}_i)$ which is unbiased even if the selection model $\hat{m}(\cdot)$ is misspecified. In other hand, if $\hat{m}(\cdot)$ is correctly specified then $\frac{\mathbb{E}[\mathbf{1}_{Revolve}]}{\mathbb{E}[\hat{m}(\mathbf{X})]} = 1$ and $ATE = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$

$\delta_{Revolve}, \delta_{Point}$ are identified under the assumption of selection on observables (Heckman, Ichimura, and Todd, 1997; Chernozhukov et al., 2018). This assumption implies that, conditional on the vector of control variables $\mathbf{X}$, treatment assignments (i.e., $\mathbf{1}_{Revolve}, \mathbf{1}_{Point}$) are as-if randomly assigned. Formally,

Assumption 3 (Unconfoundedness). $\{Y(0), Y(1)\} \perp D_i | X$

Assumption 1 asserts that controlling for all relevant covariates $\mathbf{X}$, cardholders do not revolve ($\mathbf{1}_{Revolve}$) or adopt point cards ($\mathbf{1}_{Point}$) based on unobserved factors that also influence their credit card usage intensity. Potential violation of unconfoundedness arises if time-inconsistent consumers revolve strategically to limit consumption (Bertaut, Haliassos, and Reiter, 2009). Under this scenario, difference in card usage between *Revolvers* and *Convenience users* conflates the former’s self-control motives with the true heterogeneous responsiveness to rewards, rendering the causal parameter $\delta_{Revolve}$ unidentifiable. Such possibility is unlikely in our context, as the self-control argument applies to the overall level of consumption (\$), rather than the composition of payment methods (% of consumption). For example, given a consumption level of \$2,000/month, consumers are free to allocate 50% of that expenditure toward credit cards. Controlling for demographic factors, difference in credit card usage between *Revolvers* and *Convenience users* should therefore arise from heterogeneous preferences for card features—most notably, rewards. Likewise, since cashback and point cards differ only in their reward structures, selection into these products is unlikely to be driven by other unobserved characteristics. Rather, because loyalty programs (e.g., AirMiles) offer high potential returns to affluent users via high-value redemption categories such as flight discount, selection into points cards predominantly a function of income—an observable factor I explicitly control for. Figure 3 plots the estimated causal parameters, $\delta_{Revolve}$ and δ_{Points} .

Results

While prior literature typically models credit card rewards as a homogeneous attribute (Felt et al., 2023; Agarwal et al., 2023; Wang, 2023), I document substantial heterogeneity in preferences across cardholders groups and reward structures. Results indicate that the marginal effects of rewards on credit card usage, measured as a percentage of total consumption, is 13 percentage

points (pps) lower for *Revolvers* relative to *Convenience users*. This pattern is robust to the reward types, albeit more pronounced among cardholders collecting cashback compared to those earning points (18.9 pps vs. 11.3 pps). One plausible explanation for the observed difference between revolving and non-revolving cardholders lies in the fundamental value proposition of credit cards to these groups. Whereas *Convenience users* are attracted by generous reward programs (Ching and Hayashi, 2010; Wang, 2023), *Revolvers* utilize credit cards to insure against liquidity shocks (Telyukova, 2013; Fulford and Schuh, 2024). Because reward accumulation is secondary to this consumption-smoothing motive, it follows that *Revolvers* are less sensitive to changes in reward earning rates.

In term of reward structures, consumers are slightly less responsive to changes in points than cashback ($\delta_{Points} = -2$). This effect stems primarily from *Convenience users* ($\delta_{Points}^{Conv} = -1.76$) as the estimated coefficient for the *Revolvers* subsample is insignificant. Because point earning rates are measured in cashback-equivalent terms,⁴¹ observed disparity in the marginal effects different reward types indicates structural differences in the perceived utility between cashback and points. While loyalty programs offer high potential returns to affluent users capable of utilizing high-value reward categories, the lack of a fixed value also exposes cardholders to the risk of issuers adjusting the conversion rate arbitrarily (Bureau, 2023). Furthermore, most reward points expire if an account is deemed inactive by the issuer, whereas cashback balances remain accessible insofar as the credit card is valid. Since δ_{Point} reflects the weighted average differential effects of cashback versus reward points across the income distribution, a negative CATE suggest that loyalty programs benefit high-income consumers disproportionately while offering less value to other groups.

10.3 Incremental Effects Of Credit Cards

To quantify impacts of credit cards on welfare inequality, I compare the Gini coefficient of total welfare in a counterfactual economy without credit cards with that in the observed economy with credit cards. Baseline welfare in the absence of credit cards is calculated according to:

$$W_{\text{baseline}} = \mathbf{v}_{\text{without cc}} \cdot (1 - s_d) \cdot \text{Expenditure}_{\text{with credit}}$$

⁴¹Point earning rate= points/\$ spent x Average \$ value per point

Figure 17: Caption

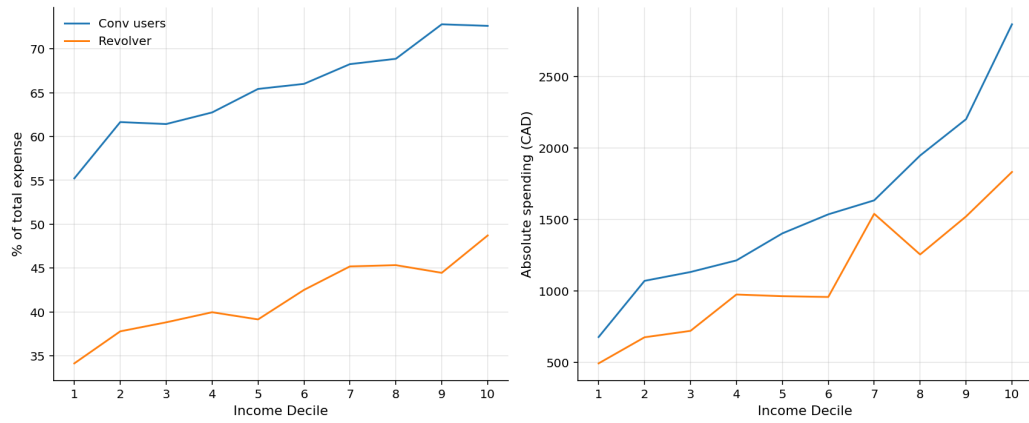


Figure 18: Heterogeneity in Preferences-Double ML

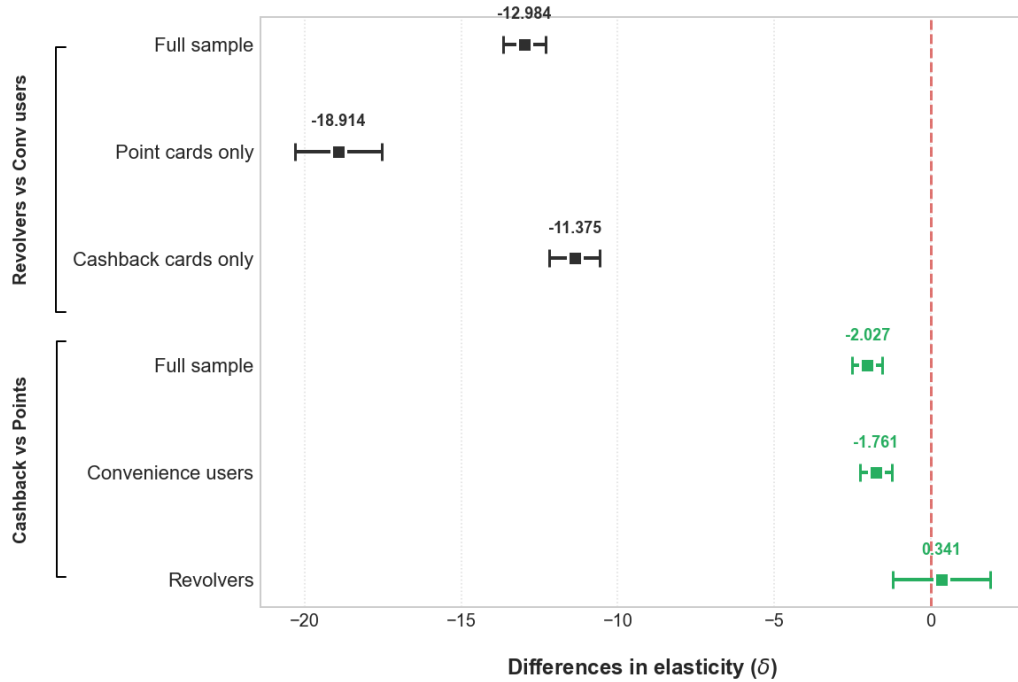
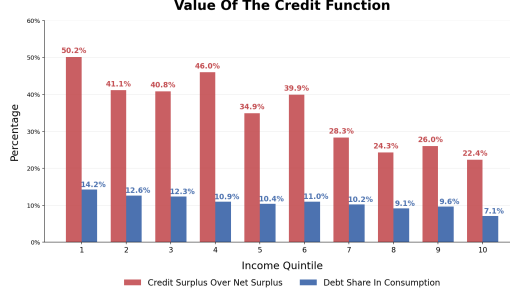


Figure 19: Caption



Since consumer surplus is defined as the net welfare gains due to the introduction of credit cards, welfare levels with credit cards are given by:

$$W_{\text{with cc}} = W_{\text{baseline}} + CS$$

The incremental effect of credit cards on welfare inequality is therefore measured as:

$$\frac{Gini_{W_{\text{with cc}}} - Gini_{W_{\text{baseline}}}}{Gini_{W_{\text{baseline}}}}$$

Since baseline welfare depends on specific calibrations of per-unit surplus without credit cards, the magnitude of the difference between the Gini coefficients is highly sensitive to specific calibrations of the average utility of cash-based methods (i.e., δ_c).

Relative to baseline calibration of δ_c specified in Eq.(14), imposing $\delta_c = 0$ would mechanically understate W_{baseline} while leaving CS largely unchanged. Estimates under this assumption therefore provide a conditional upper bound on the incremental effect of credit cards on inequality. Table 9 reports estimates under both the baseline calibration and this upper-bound specification.

Under both calibrations, total welfare remains highly unequal, with Gini coefficients ranging from 0.51 to 0.54. The estimated incremental contribution of credit cards to welfare inequality ranges from 2% under the baseline calibration to 13% under the upper-bound specification. Thus, although inequality in total welfare remains substantial, credit cards play a rather limited role in shaping this inequality.

Table 9: Incremental Effect of Credit Cards on Inequality

Notes: The baseline specification calibrates $\delta_c = \max\{0, \Xi_c - \Xi_{cc}\}$.

Gini coefficients are calculated following Raffinetti, Siletti, and Vernizzi, 2015 to accommodate for negative observations.

	Baseline calibration	$\delta_c = 0$
Welfare without credit cards, W_{baseline}	63.79	13.22
Net welfare gain from credit cards, CS	22.87	20.74
Welfare with credit cards, $W_{\text{with cc}}$	86.66	33.96
$Gini_{\text{Rewards}}$	.791	.791
$Gini_{W_{\text{baseline}}}$	.499	.479
$Gini_{W_{\text{with cc}}}$	.509	.541
Δ Gini coefficient relative to $Gini_{\text{Rewards}}$	55.4%	46.2%
Δ Gini coefficient relative to W_{baseline}	2.0%	12.9%

10.4 Additional Tables And Figures

Table 10: Determinants of Annual Fees

	Annual fees
Rewards scale	48.432*** (1.975)
No insurance	12.941*** (0.640)
Interest rate	-0.226 (0.330)
Constant	-52.403*** (7.001)
Observations	10,309
R^2	0.335
Adjusted R^2	0.334

Notes: The table reports OLS estimates from a regression of annual fees on rewards scale, insurance coverage, and the interest rate. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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Table 11: Characteristics of primary cards: *Revolvers* vs. *Convenience users*

Quintile	Avg. Reward Rate (%)		Avg. Annual Fee (\$)		Avg. Interest (%)		Avg. Limit (\$)		Avg. Utilization (%)		Avg. Insurances (#)		Share with point card	
	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.	Conv.	Rev.
1st (Bottom 20%)	1.07	0.86	33.41	29.02	20.91	20.93	12,913	12,995	56.92	56.92	3.33	3.14	0.55	0.49
2nd Quintile	1.19	0.96	45.46	45.97	21.01	20.84	17,111	17,547	49.82	49.82	3.90	3.78	0.58	0.52
3rd Quintile	1.30	1.06	60.00	52.54	20.89	20.60	20,726	20,034	32.99	32.99	4.47	4.03	0.60	0.54
4th Quintile	1.36	1.18	67.02	59.23	20.98	20.70	21,590	21,428	36.11	36.11	4.70	4.31	0.60	0.60
5th (Top 20%)	1.61	1.22	105.77	65.32	21.12	20.29	27,731	26,560	24.68	24.68	5.96	4.43	0.71	0.57

Notes: Conv. = *Convenience users*; Rev. = *Revolvers*.

Reward rates are expressed as percentages. Limit is the average total credit limit across all cards. Utilization is balance divided by limit.

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